

## Two Dimensional Yang–Mills Theory via Stochastic Differential Equations

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The equation of parallel transport is given meaning in the quantized two dimensional Euclidean Yang–Mills theory by interpreting it as a stochastic differential equation in the complete axial gauge. The expectation of products of Wilson loops is evaluated using the stochastic differential calculus and a method of Bralic. Euclidean invariance of the theory is proven. © 1989 Academic Press, Inc.

### 1. INTRODUCTION

This paper presents an analysis of quantized Euclidean Yang–Mills gauge theory in two space time dimensions. The theory is formulated in the continuum without a cutoff. This is possible because in complete axial gauge the Euclidean Yang–Mills action defines a Gaussian measure for the gauge field. The long range goal of our analysis is to explore the regularity properties of new observables called “lassos” which have been considered before in several contexts [6, 7, 10–12, 14–18, 20–22, 25–29, 31–33, 35]. In most formulations of quantized Yang–Mills theory, the Wilson loop variables (i.e., holonomy group elements) play a fundamental role. It is the expectation of products of these observables that we shall investigate in this paper in preparation for an analysis of the lassos in a later work. It is expected that the Wilson loop variables are highly singular in three and four space time dimensions and that averaging over paths will not improve the situation. In contrast there is reason to believe that the lassos will be more regular in this respect. Our main tool in this article is the stochastic differential calculus.

We first show that the differential equation for parallel translation along a smooth curve in  $R^2$  with respect to a typical gauge potential  $A$  can be interpreted as a stochastic differential equation. This relies on our use of a gauge for which the Yang–Mills action is Gaussian. We compute the expectation of products of Wilson loops using the beautiful technique of N. Bralic [11]. We show that Bralic’s method takes on its proper mathematical meaning in the context of stochastic differential calculus. It has long been recognized [23] that the expectation of Wilson loop

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variables are the proper nonabelian generalization of the Fourier transform. We believe that Bralic's method combined with stochastic calculus is the corresponding nonabelian calculational tool for Gaussian measures. In Section 3 we derive Bralic's differential equation, whose solution gives the expectation of products of Wilson loop variables. In order to make this into a useful computational technique it has been necessary to prove a commutativity theorem (Section 4) which, surprisingly (or not), requires use of the infinitesimal braid relations of knot theory. In the remaining sections we derive canonical forms for these expectations.

Two dimensional pure Yang-Mills theories have been investigated from several points of view. References [8, 11, 13, 20] evaluate explicitly the expectation of products of certain Wilson loops while [1-5, 23] study this theory in an axiomatic way. Our starting point differs from these in that we take the differential equation of parallel translation in the continuum as fundamental and interpret it as a stochastic differential equation. It is close in spirit to the recent work of B. Driver [15] who bridges the lattice and continuum theories of the previous works.

## 2. STOCHASTIC PARALLEL TRANSLATION FOR $YM_2$

We let  $G$  be a closed subgroup of  $U(N)$ . Denote by  $\mathcal{G}$  its Lie algebra. An element of  $\mathcal{G}$  is a skew Hermitian matrix and  $(S, T) := \text{trace}(T^* S)$ , where  $T^*$  is the Hermitian adjoint of  $T$ , is an inner product on  $\mathcal{G}$ . Choose an orthonormal (O.N.) basis  $\{T_a\}_1^r$  of  $\mathcal{G}$  with  $r = \text{dimension } \mathcal{G}$ . We define a  $\mathcal{G}$  valued white noise  $F$  on  $R^2$  as follows. Let  $F^a$  be independent white noises on  $R^2$  with constant density  $\beta^{-1}$ . Thus, by definition, for each Borel set  $B \subset R^2$  with finite Lebesgue measure,  $F^a(B)$  is a mean zero Gaussian random variable with

$$E(F^a(A) F^b(B)) = \beta^{-1} \delta^{ab} |A \cap B|. \quad (2.1)$$

Moreover,  $F^a(A \cup B) = F^a(A) + F^a(B)$  if  $A$  and  $B$  are disjoint. We put

$$F(B) = \sum_a F^a(B) T_a. \quad (2.2)$$

If  $S$  and  $T$  are in  $\mathcal{G}$  then (2.1) implies that

$$E((S, F(A))(T, F(B))) = \beta^{-1} (S, T) |A \cap B|, \quad S, T \text{ in } \mathcal{G}, \quad (2.3)$$

which in fact is equivalent to (2.1). It follows from (2.3) that for any orthogonal transformation  $R: \mathcal{G} \rightarrow \mathcal{G}$  the process  $B \rightarrow RF(B)$  is again a  $\mathcal{G}$  valued white noise with the same distribution as  $F$ . In particular  $F$  is invariant in distribution under the action of the adjoint representation of  $G$  on  $\mathcal{G}$ .

In order to motivate the next definition let us write, informally, for a Borel set

$B$  of finite measure,  $F(B) = \int_B F(x, y) dx dy$ . Then the white noise  $F$  is determined by the informal expression

$$d\mu(F) = Z^{-1} \exp \left[ (\beta/2) \int_{R^2} \text{trace}(F(x, y)^2) dx dy \right] \mathcal{D}F, \quad (2.4)$$

where  $\mathcal{D}F$  is an infinite dimensional "Lebesgue measure" on some space of generalized  $\mathcal{G}$  valued functions  $F$ . Proceeding informally we write

$$A_1(x, y) = \int_0^y F(x, t) dt. \quad (2.5)$$

Then

$$\partial_y A_1(x, y) = F(x, y) \quad (2.6)$$

and

$$A_1(x, 0) = 0, \quad x \text{ in } R.$$

Put  $A_2 = 0$  and  $A = A_1 dx + A_2 dy$ . Then  $A$  is a  $\mathcal{G}$  valued (generalized) 1-form in the complete axial gauge. Its curvature is  $F_{21}(x, y) = \partial_y A_1(x, y) = F(x, y)$ . Since  $A$  and  $F$  are linearly related we have, informally,  $\mathcal{D}F = \text{const } \mathcal{D}A$ . Hence (2.4) is the usual  $YM_2$  Euclidean measure. If  $\sigma: [a, b] \rightarrow R$  is  $C^1$  then parallel translation with respect to  $A$  along the curve  $x \rightarrow (x, \sigma(x))$  is defined informally by the equation

$$dU(x)/dx = -A_1(x, \sigma(x)) U(x), \quad U(a) = I \quad (2.7)$$

since the evaluation of  $A$  in the tangent direction  $(1, \sigma'(x))$  is just  $A_1$ . Here  $U(x)$  is an operator on  $\mathbb{C}^N$  and  $I$  is the identity operator. Now  $A_1$  is highly singular in view of (2.5) and therefore (2.7) needs to be interpreted carefully. To do this we put

$$M(x) = \int_a^x A_1(s, \sigma(s)) ds. \quad (2.8)$$

By (2.5) we therefore have

$$M(x) = \int_a^x \int_0^{\sigma(s)} F(s, t) dt ds$$

which is a well-defined Gaussian random variable for each  $x$  when interpreted simply as

$$M(x) = F(\text{"area under } \sigma \text{ from } a \text{ to } x").$$

More specifically, if  $B \doteq \{(s, t): a \leq s \leq x, 0 \leq t \leq \sigma(s)\}$  while  $C = \{(s, t): a \leq s \leq x, \sigma(s) \leq t < 0\}$  then we define

DEFINITION 2.1.  $M(x) = F(B) - F(C)$ .

It is clear that  $x \rightarrow M(x)$  has independent mean zero Gaussian increments and is therefore a martingale. Since  $A_1(x, \sigma(x)) = dM(x)/dx$ , Eq. (2.7) may be interpreted as a stochastic differential equation,  $dU(x) = -dM(x) U(x)$ . However, as is well known, the solution  $U(x)$  to this equation does not remain in  $G$  unless the equation is interpreted in the Fisk-Stratonowicz sense.

Let

$$C = \sum_{a=1}^r T_a^2$$

be the Casimir operator and note that Definition 2.1 implies that  $E((M^a(x + \Delta x) - M^a(x))(M^b(x + \Delta x) - M^b(x))) = \beta^{-1} \delta^{ab} |\sigma(x)| \Delta x + o(\Delta x)$  for small positive  $\Delta x$ . Hence the quadratic variation (i.e., square function) for  $M$  is

$$\langle dM^a(x) dM^b(x) \rangle = \beta^{-1} \delta^{ab} |\sigma(x)| dx. \quad (2.9)$$

LEMMA 2.2. *The solution to the Ito stochastic differential equation*

$$dU(t) = -dM(t) U(t) + \frac{1}{2} \beta^{-1} C |\sigma(t)| U(t) dt, \quad U(a) = I \quad (2.10)$$

on the interval  $[a, b]$  lies in  $G$  for each  $t$ .

*Proof.* The proof of existence and uniqueness of solutions to such stochastic differential equations is well known [19, 30, 34]. The proof that  $U(t)$  lies in  $G$  follows by a standard method [19] which we include here for the reader's convenience in this simple case. Let  $f: M_N \rightarrow R$  be a function on the set  $M_N$  of all  $N \times N$  complex matrices. Assume that as a function on  $R^{2N^2}$  it is  $C^\infty$  and has the property that  $f(gm) = f(m)$  for all  $m$  in  $M_N$  and all  $g$  in  $G$ . Write  $f'(m)\langle u \rangle = df(m + su)/ds|_{s=0}$  and  $f''(m)\langle u, v \rangle = d^2f(m + sv)\langle u \rangle/ds^2|_{s=0}$  for the first and second directional derivatives. Then for any element  $T$  in  $\mathcal{G}$  and  $m$  in  $M_N$  we have  $f(e^{sT}m) = f(m)$  for all real  $s$ . Thus  $0 = df(e^{sT}m)/ds = f'(e^{sT}m)\langle e^{sT}Tm \rangle$ . Take the derivative again at  $s=0$  to get

$$f''(m)\langle Tm, Tm \rangle + f'(m)\langle T^2m \rangle = 0. \quad (2.11)$$

The previous equation at  $s=0$  reads

$$f'(m)\langle Tm \rangle = 0. \quad (2.12)$$

By Ito's Lemma, (2.9), and (2.10) we have

$$\begin{aligned} df(U(t)m) &= f'(U(t)m)\langle dU(t)m \rangle + \frac{1}{2} f''(U(t)m)\langle dU(t)m, dU(t)m \rangle \\ &= f'(U(t)m)\langle -(dM(t)) U(t)m + \frac{1}{2} \beta^{-1} C |\sigma(t)| U(t)m dt \rangle \\ &\quad + \frac{1}{2} f''(U(t)m)\langle -(dM(t)) U(t)m, -(dM(t)) U(t)m \rangle \end{aligned}$$

$$\begin{aligned}
&= \sum_{a=1}^r -f'(U(t)m) \langle T_a U(t)m \rangle dM^a(t) \\
&\quad + \frac{1}{2}\beta^{-1} |\sigma(t)| \sum_a f'(U(t)m) \langle T_a^2 U(t)m \rangle dt \\
&\quad + \frac{1}{2}\beta^{-1} |\sigma(t)| \sum_a f''(U(t)m) \langle T_a U(t)m, T_a U(t)m \rangle dt.
\end{aligned}$$

The first line is zero by (2.12). The second and third lines add to zero by (2.11). Hence  $df(U(t)m) = 0$ . Thus  $f(U(t)m) = f(m)$  since  $U(0) = I$ . Choose for  $f$  a function of compact support which is one exactly on  $G$ . For example if  $h$  is a  $C^\infty$  function on  $GL(N, \mathbb{C})/G$  with compact support such that  $h = 1$  exactly on the image of  $I$  in the quotient space, then its pullback to  $GL(N, \mathbb{C})$  is one exactly on  $G$  and has compact support in  $GL(N, \mathbb{C})$ . We may take  $f$  to be this pullback extended to be zero off  $GL(N, \mathbb{C})$ . Putting  $m = I$  we therefore have  $f(U(t)) = f(I) = 1$  for all  $t$  in  $[a, b]$ . So  $U(t)$  is in  $G$ .

### 3. BRALIC'S DIFFERENTIAL EQUATION

We consider  $2k$  curves in  $R^2$  lying over an interval  $[c, d]$  of the  $x$ -axis. Each curve is the graph of a piecewise  $C^1$  function  $\sigma_j: [c, d] \rightarrow R$ ,  $j = 1, \dots, 2k$ . Define  $M_j(x)$  for  $\sigma_j$  just as  $M(x)$  was defined for  $\sigma$  in Section 2 and write  $U_j(x)$  for the corresponding stochastic parallel translation operator along  $\sigma_j$ . Writing  $M_j(x) = \sum_{a=1}^r M_j^a(x) T_a$  as before it follows from (2.1) and Definition 2.1 that the quadratic covariations of these martingales are given by

$$\langle dM_i^a(x) dM_j^b(x) \rangle = \beta^{-1} \delta^{ab} m_{ij}(x), \quad 1 \leq i, j \leq 2k, \quad (3.1)$$

where

$$m_{ij}(x) = \begin{cases} \min\{|\sigma_i(x)|, |\sigma_j(x)|\} & \text{if } \sigma_i(x) \sigma_j(x) \geq 0 \\ 0 & \text{if } \sigma_i(x) \sigma_j(x) < 0. \end{cases} \quad (3.2)$$

Consider the  $k$ -fold tensor product representation of  $G$ . Put  $g^{(k)} = g \otimes \dots \otimes g$  ( $k$  factors) operating on the  $k$ -fold tensor product  $(\mathbb{C}^N)^k$ . Denote by  $\mathcal{B}$  the commutant algebra. That is,

$$\mathcal{B} = \{B: (\mathbb{C}^N)^k \rightarrow (\mathbb{C}^N)^k; g^{(k)}B = Bg^{(k)} \text{ for all } g \text{ in } G\}. \quad (3.3)$$

If  $P$  is in the group  $S_k$  of permutations of  $\{1, \dots, k\}$  then the operator  $\hat{P}$  defined by

$$\hat{P}(x_1 \otimes \dots \otimes x_k) = x_{P(1)} \otimes \dots \otimes x_{P(k)} \quad (3.4)$$

is clearly in  $\mathcal{B}$ .

We define a map  $\varphi(c, x)$  on  $\mathcal{B}$  by

$$\begin{aligned} \varphi(c, x)B &= E[(U_{k+1}(x)^{-1} \otimes \cdots \otimes U_{2k}(x)^{-1}) \\ &\quad \times B(U_1(x) \otimes \cdots \otimes U_k(x))] \end{aligned} \quad (3.5)$$

for  $B$  in  $\mathcal{B}$  and  $c \leq x \leq d$ . We call  $\varphi(c, x)$  Bralic's map for the interval  $[c, x]$ .

LEMMA 3.1.  $\varphi(c, x)B$  is in  $\mathcal{B}$ .

*Proof.* We shall use global gauge invariance. Fix  $g$  in  $G$ . Put  $U_j^g(x) = gU_j(x)g^{-1}$ ,  $F^g(A) = gF(A)g^{-1}$  for any Borel set  $A \subset R^2$  and put  $M_j^g(x) = gM_j(x)g^{-1}$ . Multiplying (2.10) on the left by  $g$  and on the right by  $g^{-1}$  we see that  $U^g$  is the solution to (2.10) determined by the martingale  $M^g$  since  $gC = Cg$ . Thus  $U_j^g(x)$  is the parallel translation operator induced by the white noise  $F^g$ . As already noted  $F$  and  $F^g$  have the same distributions. Hence the right side of (3.5) is equal to

$$E[(U_{k+1}^g(x) \otimes \cdots \otimes U_{2k}^g(x))^{-1} B(U_1^g(x) \otimes \cdots \otimes U_k^g(x))]$$

which equals

$$E[g^k(U_{k+1}(x) \otimes \cdots \otimes U_{2k}(x))^{-1} (g^{-1})^k Bg^k(U_1(x) \otimes \cdots \otimes U_k(x)) g^{-k}].$$

Since  $(g^{-1})^{(k)} Bg^{(k)} = B$  we have  $\varphi(c, x)B = g^{(k)}\varphi(c, x)Bg^{-(k)}$ . Hence  $\varphi(c, x)B$  is in  $\mathcal{B}$ .

For any operator  $T$  on  $\mathbb{C}^N$  we write

$$T^{(j)} = I \otimes \cdots \otimes I \otimes T \otimes I \otimes \cdots \otimes I \quad (3.6)$$

on  $(\mathbb{C}^N)^k$ , where the non identity factor  $T$  operates on the  $j$ th factor. For our basis  $\{T_a\}$  of  $\mathcal{G}$  we write  $T_a^{(j)}$  for  $(T_a)^{(j)}$ . Note that  $T^{(j)}$  always commutes with  $T^{(i)}$ . For any operator  $B$  on  $(\mathbb{C}^n)^k$  we define

$$S_{i,j}B = \sum_a BT_a^{(i)}T_a^{(j)}, \quad 1 \leq i, j \leq k \quad (3.7)$$

$$S_{k+i, k+j}B = \sum_a T_a^{(i)}T_a^{(j)}B, \quad 1 \leq i, j \leq k \quad (3.8)$$

$$S_{k+j, i}B = S_{i, k+j}B = -\sum_a T_a^{(j)}BT_a^{(i)}, \quad 1 \leq i, j \leq k. \quad (3.9)$$

Clearly  $S_{i,j} = S_{j,i}$  for  $1 \leq i, j \leq 2k$ .

LEMMA 3.2.  $S_{i,j}$  takes  $\mathcal{B}$  into  $\mathcal{B}$  for  $1 \leq i, j \leq 2k$ . Moreover for  $B$  in  $\mathcal{B}$

$$\sum_{j=1}^{2k} S_{ij}B = 0, \quad i = 1, \dots, 2k. \quad (3.10)$$

*Proof.* Since the adjoint representation of  $G$  on  $\mathcal{G}$  is an orthogonal action there

is an orthogonal matrix  $\{g_{ab}\}$  such that  $g^{-1}T_a g = \sum_b g_{ba}T_b$ . Hence for  $B$  in  $\mathcal{B}$  and  $1 \leq i, j \leq k$  we have, using summation convention,  $(g^{-1})^k S_{i,j} B g^{(k)} = (g^{-1})^{(k)} B g^{(k)} (g^{-1})^k T_a^{(i)} T_a^{(j)} g^{(k)} = B (g^{-1}T_a g)^{(i)} (g^{-1}T_a g)^{(j)} = B g_{ba} g_{ca} T_b^{(i)} T_c^{(j)} = B T_b^{(i)} T_b^{(j)} = S_{i,j} B$ . So  $S_{i,j} B$  is in  $\mathcal{B}$ . A similar proof applies to the other ranges of  $i, j$ .

To prove (3.10) observe that if  $T$  is in  $\mathcal{G}$  and  $B$  is in  $\mathcal{B}$  then  $(e^{sT})^{(k)} B = B (e^{sT})^{(k)}$  for all real  $s$ . Differentiating this at  $s=0$  gives

$$\left[ \sum_{j=1}^k T^{(j)}, B \right] = 0. \quad (3.11)$$

Now suppose  $1 \leq i \leq k$ . Then

$$\begin{aligned} \sum_{j=1}^{2k} S_{i,j} B &= \sum_{j=1}^k \sum_a B T_a^{(i)} T_a^{(j)} - \sum_{j=1}^k \sum_a T_a^{(j)} B T_a^{(i)} \\ &= \sum_a \left[ B, \sum_{j=1}^k T_a^{(j)} \right] T_a^{(i)} \\ &= 0. \end{aligned}$$

A similar proof of (3.10) holds in case  $k+1 \leq i \leq 2k$ .

Now we define an operator  $H(x): \mathcal{B} \rightarrow \mathcal{B}$  by

$$H(x) = -(2\beta)^{-1} \sum_{i,j=1}^{2k} m_{ij}(x) S_{i,j}. \quad (3.12)$$

**THEOREM 3.3** (Bralic's differential equation).  $\varphi(c, x)$  satisfies the differential equation

$$d\varphi(c, x)/dx = -\varphi(c, x) H(x) \quad \text{on } [c, d] \quad (3.13)$$

and the initial condition  $\varphi(c, c) = I$ .

*Proof.* Using the product rule  $d(XY) = X dY + Y dX + \langle dX dY \rangle$ , for stochastic differentials, one has by induction the multivariable product rule for real valued semimartingales  $X_1, \dots, X_n$ :

$$\begin{aligned} d(X_1 \cdots X_n) &= \sum_{i=1}^n X_1 X_2 \cdots (dX_i) X_{i+1} \cdots X_n \\ &\quad + \sum_{i < j} X_1 \cdots (dX_i) X_{i+1} \cdots (dX_j) X_{j+1} \cdots X_n. \end{aligned} \quad (3.14)$$

We have omitted here the quadratic covariation bracket  $\langle \rangle$  in the second term. But it should be understood in this and the following equations. If the semimartingales  $X_1, \dots, X_n$  take their values in the space of operators on some finite dimensional vector space then by choosing a basis for operators one sees from

multilinearity that (3.14) holds if some of the products in  $X_1, \dots, X_n$  are operator products and some are tensor products. Since the Casimir operator  $C$  is Hermitian and  $U(x)$  is unitary we may take the adjoint of Eq. (2.10) to get

$$dU(x)^{-1} = U(x)^{-1} dM(x) + \frac{1}{2}\beta^{-1}CU(x)^{-1}|\sigma(x)|dx, \quad (3.15)$$

where we have used  $M(x)^* = -M(x)$ . We put  $V(x) = U_{k+1}(x)^{-1} \otimes \dots \otimes U_{2k}(x)^{-1}$  and  $W(x) = U_1(x) \otimes \dots \otimes U_k(x)$ , and we define the following Ito stochastic differentials:

$$\begin{aligned} V_i(x) &= U_{k+1}(x)^{-1} \otimes \dots \otimes (dU_{k+i}(x)^{-1}) \otimes \dots \otimes U_{2k}(x)^{-1}, & 1 \leq i \leq k \\ W_i(x) &= U_1(x) \otimes \dots \otimes (dU_i(x)) \otimes \dots \otimes U_k(x), & 1 \leq i \leq k \\ V_{i,j}(x) &= U_{k+1}(x)^{-1} \otimes \dots \otimes (dU_{k+i}(x)^{-1}) \\ &\quad \otimes \dots \otimes (dU_{k+j}(x)^{-1}) \otimes \dots \otimes U_{2k}(x)^{-1}, & 1 \leq i < j \leq k \\ W_{i,j}(x) &= U_1(x) \otimes \dots \otimes (dU_i(x)) \\ &\quad \otimes \dots \otimes (dU_j(x)) \otimes \dots \otimes U_k(x), & 1 \leq i < j \leq k. \end{aligned}$$

Then by (3.14) we have for  $B$  in  $\mathcal{B}$

$$\begin{aligned} d(V(x)BW(x)) &= \sum_{i=1}^k (V_i(x)BW(x) + V(x)BW_i(x)) \\ &\quad + \sum_{i,j=1}^k V_j(x)BW_i(x) \\ &\quad + \sum_{1 \leq i < j \leq k} (V_{i,j}(x)BW(x) + V(x)BW_{i,j}(x)). \end{aligned} \quad (3.16)$$

Let us now compute the quadratic differentials. By (2.10) and (3.1) and the usual rules of the stochastic calculus we have

$$\begin{aligned} dU_i(x) \otimes dU_j(x) &= -(dM_i(x))U_i(x) \otimes (-dM_j(x))U_j(x) \\ &= \sum_{a,b} T_a U_i(x) \otimes T_b U_j(x) \langle dM_i^a(x) dM_j^b(x) \rangle \\ &= \beta^{-1} \left( \sum_a T_a \otimes T_a \right) m_{ij}(x) U_i(x) \otimes U_j(x) dx. \end{aligned}$$

Placing  $U_i(x)$  and  $U_j(x)$  in their correct positions in  $W(x)$  we therefore have, with  $T_{ij} := \sum_a T_a^{(i)} T_a^{(j)}$ .

$$W_{i,j}(x) = \beta^{-1} m_{ij}(x) T_{ij} W(x) dx, \quad 1 \leq i < j \leq k. \quad (3.17)$$

Similarly

$$V_{i,j}(x) = \beta^{-1} m_{k+i,k+j}(x) V(x) T_{ij} dx, \quad 1 \leq i < j \leq k \quad (3.18)$$



and

$$V_j(x) BW_i(x) = \beta^{-1} m_{i,k+j}(x) V(x)(S_{i,k+j}B) W(x) dx, \quad 1 \leq i, j \leq k. \quad (3.19)$$

Hence the second and third terms in (3.16) add to

$$\beta^{-1} \sum_{1 \leq r < s \leq 2k} m_{r,s}(x) V(x)(S_{r,s}B) W(x) dx.$$

Now the first term on the right of (2.10) has expectation zero since  $M$  is a martingale. Hence

$$\begin{aligned} E(V(x) BW_i(x)) &= \frac{1}{2}\beta^{-1} |\sigma_i(x)| E(V(x) B(I \otimes \dots \otimes C^{(i)} \otimes \dots \otimes I) W(x)) dx \\ &= (2\beta)^{-1} |\sigma_i(x)| E(V(x)(S_{i,i}B) W(x)) dx \end{aligned}$$

and similarly

$$E(V_i(x) BW(x)) = (2\beta)^{-1} |\sigma_{k+i}(x)| E(V(x)(S_{k+i,k+i}B) W(x)) dx.$$

Thus

$$\begin{aligned} d\varphi(c, x)B &= E[d(V(x) BW(x))] \\ &= (2\beta)^{-1} \sum_{r=1}^{2k} |\sigma_r(x)| \varphi(c, x) S_{r,r}B dx \\ &\quad + E \left[ V(x) \beta^{-1} \sum_{1 \leq r < s \leq 2k} m_{r,s}(x)(S_{r,s}B) W(x) \right] dx \\ &= -\varphi(c, x) H(x)B dx. \end{aligned}$$

This proves (3.13). Since  $U_j(c) = I$  for all  $j$  we have  $\varphi(c, c)B = B$ .

**REMARK 3.4.** Since  $\varphi(c, x)$  satisfies a first order differential equation with identity initial condition we have

$$\varphi(x, y) \varphi(y, z) = \varphi(x, z), \quad c \leq x \leq y \leq z \leq d. \quad (3.20)$$

This also follows directly from the definition (3.5) because the matrix elements  $U_{\alpha\beta}$  of a parallel translation operator along a curve lying over an interval  $(x, y)$  are independent of the matrix elements of the parallel translation operator for a curve lying over a disjoint interval  $(y, z)$ . This in turn is a consequence of the fact that the increments  $dM(t)$  in (2.10) are independent.

We shall explain now the connection between Bralic's map  $\varphi(c, d)$  and the expectation of Wilson loops constructed from the  $2k$  curves  $\sigma_1, \dots, \sigma_{2k}$ .

A cycle of length  $r \geq 1$  is a permutation  $C$  of  $\{1, \dots, k\}$  of the form

$$Ci_s = \begin{cases} i_{s+1}, & s = 1, \dots, r-1 \\ i_1, & s = r \end{cases} \quad (3.21)$$

while  $Cn = n$  if  $n$  is not in the set  $|C| := \{i_1, \dots, i_r\}$ . Any permutation of  $\{1, \dots, k\}$  is uniquely a product of disjoint cycles;

$$P = C_1 \cdots C_m, \quad (3.22)$$

where  $|C_i| \cap |C_j| = \emptyset$  if  $i \neq j$  and wherein we adopt the convention that  $\bigcup_{i=1}^m |C_i| = \{1, \dots, k\}$ .

If  $A_1, \dots, A_k$  are operators on  $\mathbb{C}^N$  we write

$$A^C = A_{i_1} A_{i_2} \cdots A_{i_r}. \quad (3.23)$$

LEMMA 3.5. If  $A_1, \dots, A_k$  are operators on  $\mathbb{C}^N$  and  $P$  is given by (3.22) then

$$\text{trace}[(A_1 \otimes \cdots \otimes A_k) \hat{P}] = \prod_{j=1}^m \text{trace}[A^{C_j}]. \quad (3.24)$$

*Proof.* Consider first the case in which  $P$  consists of just one cycle  $(k, \dots, 2, 1)$ , which takes  $j$  to  $j+1$  for  $j = 1, \dots, k-1$ . If  $\{e_\alpha\}$  is an O.N. basis for  $\mathbb{C}^N$  then

$$\begin{aligned} & \text{trace}[(A_1 \otimes \cdots \otimes A_k) \hat{P}] \\ &= \sum_{\alpha_1, \dots, \alpha_k} ((A_1 \otimes \cdots \otimes A_k) \hat{P} e_{\alpha_1} \otimes \cdots \otimes e_{\alpha_k}, e_{\alpha_1} \otimes \cdots \otimes e_{\alpha_k}) \\ &= \sum ((A_1 \otimes \cdots \otimes A_k)(e_{\alpha_2} \otimes \cdots \otimes e_{\alpha_k} \otimes e_{\alpha_1}), e_{\alpha_1} \otimes \cdots \otimes e_{\alpha_k}) \\ &= \sum (A_1 e_{\alpha_2}, e_{\alpha_1})(A_2 e_{\alpha_3}, e_{\alpha_2}) \cdots (A_k e_{\alpha_1}, e_{\alpha_k}) \\ &= \sum (A_1)_{\alpha_1 \alpha_2} (A_2)_{\alpha_2 \alpha_3} \cdots (A_k)_{\alpha_k \alpha_1} \\ &= \text{trace}[A_1 A_2 \cdots A_k]. \end{aligned}$$

We reduce the general case to the preceding case as follows. Let  $r_j$  be the length of  $C_j$  and put  $k_j = r_1 + \cdots + r_{j-1}$  if  $j \geq 2$  and put  $k_1 = 0$ . Write  $C_j = (i_{r_j}^j, \dots, i_2^j, i_1^j)$  and put  $R(k_j + s) = i_s^j$  for  $s = 1, \dots, r_j$ ,  $j = 1, \dots, m$ . Then  $D_j := R^{-1} C_j R = (k_j + r_j, \dots, k_j + 1)$ . Put  $B_i = A_{R(i)}$  for  $i = 1, \dots, k$  and  $B^j = (B_{k_j+1} \otimes B_{k_j+2} \otimes \cdots \otimes B_{k_j+r_j})(r_j, \dots, 1)^\wedge$  on  $(\mathbb{C}^N)^{r_j}$ . Then

$$\begin{aligned}
\text{trace}[(A_1 \otimes \cdots \otimes A_k) \hat{P}] &= \text{trace}[\hat{R}(A_1 \otimes \cdots \otimes A_k) \hat{R}^{-1}(R^{-1}PR)^\wedge] \\
&= \text{trace}[(B_1 \otimes \cdots \otimes B_k) \hat{D}_1 \cdots \hat{D}_m] \\
&= \text{trace}[B^1 \otimes \cdots \otimes B^m] \\
&= \prod_{j=1}^m \text{trace}[B^j] \\
&= \prod_{j=1}^m \text{trace}[B_{k_j+1} \cdots B_{k_j+r_j}] \\
&= \prod_{j=1}^m \text{trace}[A^{C_j}].
\end{aligned}$$

DEFINITION 3.6. Let  $U_1, \dots, U_k, V_1, \dots, V_k$  be operators on  $\mathbb{C}^N$ . Let  $P$  and  $Q$  be permutations in  $S_k$ . An operator loop for  $P, Q$  is a product of the form

$$V_i U_{P(i)} V_{QP(i)} U_{PQP(i)} \cdots V_{(QP)^{r-1}(i)} U_{P(QP)^{r-1}(i)}, \quad (3.25)$$

where  $(QP)^r(i) = i$  and  $r$  is the length of some cycle in the cycle decomposition of  $QP$ .

EXAMPLE 3.7. For  $k=2$  and  $P=Q=I$  there are two operator loops, namely,  $V_1 U_1$  and  $V_2 U_2$ . For  $k=2, P=\tau_{12}, Q=I$  there is only one operator loop up to cyclic permutation, namely,  $V_1 U_2 V_2 U_1$ .

Remark 3.8. Two operator loops either coincide after a cyclic permutation or else no operator  $U_i$  or  $V_j$  that occurs in one product occurs in the other. (The operators  $U_1, \dots, V_k$  are to be regarded as distinct in this statement.) We say two operator loops are *distinct* if one cannot be changed into the other by a cyclic permutation. The distinct operator loops are therefore in one-to-one correspondence with the cycles in  $QP$ .

Note that each operator  $V_j$  that occurs in an operator loop is preceded on the right by the operator  $U_{P(j)}$  and each operator  $U_j$  in the operator loop is preceded on the right by  $V_{Q(j)}$ , if one interprets this in the cyclic sense that the operator on the far left precedes the operator on the far right.

COROLLARY 3.9. Let  $U_1, \dots, U_k, V_1, \dots, V_k$  be operators on  $\mathbb{C}^N$  and let  $P$  and  $Q$  be in  $S_k$ . Then

$$\text{trace}[\hat{Q}(V_1 \otimes \cdots \otimes V_k) \hat{P}(U_1 \otimes \cdots \otimes U_k)] = \prod_{j=1}^m \text{trace}[W_j], \quad (3.26)$$

where  $W_1, \dots, W_m$  are the distinct operator loops for  $P, Q$ .

*Proof.* The left side of (3.26) equals

$$\begin{aligned} & \text{trace}[(V_1 \otimes \cdots \otimes V_k)(U_{P(1)} \otimes \cdots \otimes U_{P(k)}) \hat{P} \hat{Q}] \\ &= \text{trace}[(V_1 U_{P(1)}) \otimes \cdots \otimes (V_k U_{P(k)})](QP)^{\wedge}. \end{aligned}$$

Put  $A_i = V_i U_{P(i)}$ ,  $i = 1, \dots, k$  and apply Lemma 3.5 to  $A_1, \dots, A_k$  and to the cycle decomposition of  $QP$ . Since a typical  $A^C$  is of the form  $A_i A_{(QP)(i)} \cdots A_{(QP)^{r-1}(i)}$  we see from (3.25) that the operators  $A^C$  are just the operator loops  $W_j$ . Thus (3.24) reduces to (3.26).

To obtain the expectation of Wilson loops from Bralic's map consider the  $2k$  curves of Theorem 3.3. Write  $U_j$  for the parallel translation operator along the graph of  $\sigma_j$  from  $c$  to  $d$ ,  $j = 1, \dots, 2k$ . Then  $V_j := U_{k+j}^{-1}$  is a parallel translation from  $(d, \gamma_j(d))$  to  $(c, \gamma_j(c))$  (i.e., from right to left) along the graph of  $\gamma_j := \sigma_{k+j}$ . Now let  $P$  and  $Q$  be permutations in  $S_k$  and let  $C_j$  be a cycle in  $QP$  beginning with  $i$ . So  $(QP)^r(i) = i$  if  $r$  is the length of  $C_j$ . We denote by  $\tau_j$  the curve (which is the graph of)  $\gamma_i$  oriented from right to left preceded by the vertical segment from  $\sigma_{P(i)}(d)$  to  $\gamma_i(d)$  and then preceded by the curve  $\sigma_{P(i)}$  oriented from left to right and then preceded by the vertical segment from  $\gamma_{QP(i)}(c)$  to  $\sigma_{P(i)}(c)$  and then preceded by  $\gamma_{QP(i)}$  oriented from right to left, etc., until we arrive finally at  $(d, \gamma_i(d))$  again. If  $U(\tau_j)$  denotes parallel translation around the closed curve  $\tau_j$  then it is clear from its construction that  $U(\tau_j)$  is exactly an operator loop for  $P, Q$  since parallel translation along a vertical line segment is the identity operator. In this way each cycle in the decomposition of  $QP$  gives a closed curve consisting of the graphs of some of the  $\sigma_s$ ,  $s = 1, \dots, k$  oriented toward the right interspersed with the graphs of some of the  $\sigma_{k+s}$ ,  $s = 1, \dots, k$  oriented toward the left. The two sets of curves are connected by vertical line segments at the two endpoints in accordance with instructions determined by the permutation  $P$  (for connections at the right endpoint) and  $Q$  (for connections at the left endpoint). By Corollary 3.6 we have

$$\text{trace}[\hat{Q}(V_1 \otimes \cdots \otimes V_k) \hat{P}(U_1 \otimes \cdots \otimes U_k)] = \prod_{j=1}^m \text{trace}[U(\tau_j)]. \quad (3.27)$$

In view of the Definition 3.5 we may now evaluate the expectations on the right of (3.27) in terms of Bralic's map  $\varphi(c, d)$  since  $\hat{P}$  is in  $\mathcal{B}$ . We summarize in the following theorem.

**THEOREM 3.10.** *Let  $\sigma_1, \dots, \sigma_{2k}$  be continuous real valued functions on  $[c, d]$ . Let  $P$  and  $Q$  be permutations in  $S_k$ . Denote by  $\tau_1, \dots, \tau_m$  the closed curves constructed above from  $\sigma_1, \dots, \sigma_{2k}$ ,  $P, Q$  and by  $U(\tau_j)$  parallel translation around  $\tau_j$ . Then*

$$E\left(\prod_{j=1}^m \text{trace}[U(\tau_j)]\right) = \text{trace}[\hat{Q}\varphi(c, d)\hat{P}]. \quad (3.28)$$

*Remark 3.11.* Although only the permutations  $\hat{P}$  in  $\mathcal{B}$  are needed for evaluation

of Wilson loops as in (3.28) nevertheless we must consider the action of  $\varphi(c, x)$  on all of  $\mathcal{B}$  because it is this algebra which is invariant under  $\varphi(c, x)$ . (Use of a larger algebra would not permit a derivation of some important identities of the next section.)

#### 4. LOCAL COMMUTATIVITY

We shall prove in this section that the operators  $H(x)$  and  $H(x')$  commute if the graphs of the functions  $\sigma_1, \dots, \sigma_{2n}$  do not cross between  $x$  and  $x'$ . We put  $\beta = 1$  henceforth. Equations (4.1) and (4.2) of the next Lemmas constitute the infinitesimal braid relations of knot theory, which was studied in the context of a complex simple group in [9, 24].

**LEMMA 4.1.** *Consider three representations of the Lie algebra  $\mathcal{G}$ . Denote by  $M_a, N_a, P_a$  the respective images of  $T_a$  in the three representations. Put  $R_{12} = \sum_a M_a \otimes N_a \otimes I$ ,  $R_{13} = \sum_a M_a \otimes I \otimes P_a$  and  $R_{23} = \sum_a I \otimes N_a \otimes P_a$ . Then*

$$[R_{12}, R_{13} + R_{23}] = 0. \quad (4.1)$$

*Proof.* Define structure constants by  $[T_a, T_b] = \sum_c f_{ab}^c T_c$ . Since the map  $T \rightarrow [T, T_a]$  is a skew symmetric operator on  $\mathcal{G}$  relative to the trace inner product and  $\{T_a\}_{a=1}^r$  is an O.N. basis of  $\mathcal{G}$  the structure constants satisfy  $f_{ab}^c = -f_{cb}^a$ . Hence

$$\begin{aligned} [[R_{12}, R_{13} + R_{23}]] &= \sum_{a,b} [M_a \otimes N_a \otimes I, M_b \otimes I \otimes P_b + I \otimes N_b \otimes P_b] \\ &= \sum_{ab} ([M_a, M_b] \otimes N_a \otimes P_b + M_a \otimes [N_a, N_b] \otimes P_b) \\ &= \sum_{abc} (f_{ab}^c M_c \otimes N_a \otimes P_b + f_{ab}^c M_a \otimes N_c \otimes P_b) \\ &= \sum_{abc} (f_{ab}^c + f_{cb}^a) M_c \otimes N_a \otimes P_b \\ &= 0. \end{aligned}$$

**LEMMA 4.2.** *The operators  $S_{ij}$  defined in (3.7), (3.8), and (3.9) satisfy the identities*

$$[S_{ij}, S_{im} + S_{jm}] = 0, \quad i, j, m \text{ distinct in } \{1, \dots, 2k\}. \quad (4.2)$$

*Proof.* By Lemma 4.1 with  $M_a = N_a = P_a = T_a$  the operators  $T_{ij} = \sum_a T_a^{(i)} T_a^{(j)}$ ,  $1 \leq i < j \leq k$  satisfy the identities  $[T_{ij}, T_{im} + T_{jm}] = 0$  for distinct  $i, j, m$  in  $\{1, \dots, k\}$  since the extra factors of  $I$  in  $T_{ij}$  don't affect (4.1) nor does the order of the factor spaces. Hence, by (3.7) and (3.8), Eq. (4.2) holds if  $i, j, m$  all lie in  $\{1, \dots, k\}$  or all

lie in  $\{k+1, \dots, 2k\}$ . In case  $i \leq k$  and  $m \geq k+1$  then two of the three operators in (4.2) are of the form (3.9). Consider the case  $1 \leq i < j \leq k$  and  $m \geq k+1$ . Put  $M_a = N_a = T_a$  in Lemma 4.1 and  $P_a = -T'_a$  where prime denotes transpose. Thus  $P_a$  is the contragredient representation of  $\mathcal{G}$  on the dual space  $(\mathbb{C}^N)^*$ . Upon inserting  $2k-3$  factors of the identity, (4.1) reads

$$\left[ \sum_a T_a^{(i)} T_a^{(j)}, \sum_b T_b^{(i)} (-T_b')^{(m)} + \sum_b T_b^{(j)} (-T_b')^{(m)} \right] = 0 \quad (4.3)$$

wherein each operator operates on  $(\mathbb{C}^N)^k \otimes ((\mathbb{C}^N)^*)^k$ . But for any finite dimensional vector space  $V$  the space  $V \otimes V^*$  is linearly isomorphic to the space  $\mathcal{L}(V)$  of linear operators on  $V$  under the linear extension of  $v \otimes f \rightarrow (u \rightarrow vf(u))$ . Moreover a pair of operators  $R, S$  on  $V$  determines a map  $B \rightarrow RBS$  from  $\mathcal{L}(V)$  into itself which under the isomorphism corresponds to  $R \otimes S'$ , as one verifies easily. Put  $V = (\mathbb{C}^N)^k$  and apply this isomorphism to (4.3). By the definitions of  $S_{i,j}$  the identity (4.3) then is equivalent to (4.2). The proof for  $i \leq k, m > j \geq k+1$  is similar. One applies Lemma 4.1 with  $M_a = T_a$  and  $N_a = P_a = -T'_a$  this time. And finally the case  $i \leq k, m \leq k, j \geq k+1$  follows by using  $M_a = P_a = T_a, N_a = -T'_a$ .

**THEOREM 4.3.** *Fix  $x$  and  $x'$  in  $[c, d]$  and assume that for any  $i$  and  $j$  either  $\sigma_i(t) \geq \sigma_j(t)$  for  $t = x$  and  $x'$  or  $\sigma_i(t) \leq \sigma_j(t)$  for  $t = x$  and  $x'$ . Then  $H(x)$  and  $H(x')$  commute. In particular, if the functions  $\sigma_1, \dots, \sigma_{2k}$  do not cross over  $[c, d]$  in the sense that for any  $i$  and  $j$  either  $\sigma_i(x) \geq \sigma_j(x)$  for all  $x$  in  $[c, d]$  or  $\sigma_i(x) \leq \sigma_j(x)$  for all  $x$  in  $[c, d]$  then  $H(x)$  and  $H(x')$  commute for all  $x$  and  $x'$  in  $[c, d]$ .*

*Proof.* Fix  $x$  and  $x'$  as above and choose a permutation  $i_1, \dots, i_{2k}$  so that  $\sigma_{i_n}(t) \leq \sigma_{i_{n+1}}(t)$  for  $n = 1, \dots, 2k-1$  and  $t = x$  and  $x'$ . For example choose  $i_1$  in  $\{i : \sigma_i(x) = \inf_j \sigma_j(x)\} \cap \{i : \sigma_i(x') = \inf_j \sigma_j(x')\}$  which cannot be empty without violating the hypothesis.  $i_2$  may be chosen similarly after removing  $\sigma_{i_1}$  from the procedure.

Write  $y_r = \sigma_{i_r}(x)$ . Since  $m_{ij}(x)$  and  $S_{ij}$  are both symmetric in  $i$  and  $j$  we may write

$$-2H(x) = \sum_{r=1}^{2k} m_{i_r, i_r}(x) S_{i_r, i_r} + 2 \sum_{r=1}^{2k-1} \sum_{p>r} m_{i_r, i_p}(x) S_{i_r, i_p}. \quad (4.4)$$

Let  $r_0 = \min\{r : y_r \geq 0\}$ . Then by (3.2)  $m_{i_r, i_p}(x) = 0$  unless  $r$  and  $p$  are both greater or equal to  $r_0$  or both strictly less than  $r_0$ . In the first instance  $m_{i_r, i_p} = y_r$  if  $r \leq p$  and in the second instance  $m_{i_r, i_p} = |y_p| = -y_p$  if  $r \leq p$ . Thus using  $S_{ij} = S_{ji}$  and (3.10) we have

$$\begin{aligned} 2 \sum_{r_0 > p > r} m_{i_r, i_p}(x) S_{i_r, i_p} &= -2 \sum_{r_0 > p > r} y_p S_{i_r, i_p} \\ &= -2 \sum_{r_0 > r > p} y_r S_{i_r, i_p} \\ &= +2 \sum_{r_0 > r} y_r \sum_{p \geq r} S_{i_r, i_p}. \end{aligned}$$

Hence

$$\begin{aligned} -2H(x) &= \sum_{r \geq r_0} y_r S_{i_r i_r} - \sum_{r < r_0} y_r S_{i_r i_r} + 2 \sum_{r \geq r_0} \sum_{p > r} y_r S_{i_r i_p} \\ &\quad + 2 \sum_{r < r_0} y_r \sum_{p \geq r} S_{i_r i_p} \\ &= \sum_{r=1}^{2k} y_r S_{i_r i_r} + 2 \sum_{r=1}^{2k-1} y_r \sum_{p > r} S_{i_r i_p}. \end{aligned}$$

Put

$$-H_r = \left(\frac{1}{2}\right) S_{i_r i_r} + \sum_{p > r} S_{i_r i_p}. \quad (4.5)$$

Then

$$H(x) = \sum_{r=1}^{2k} y_r H_r. \quad (4.6)$$

Similarly  $H(x')$  is a linear combination of the same operators  $H_r$  since the ordinates  $\sigma_{i_r}(x')$  are also nondecreasing. It suffices therefore to show that the operators  $H_r$  commute.

Since each  $T_a$  commutes with the Casimir operator  $C$  it follows that  $[S_{i_r i_r}, S_{i_r i_p}] = 0$  for all  $i, r, p$ . Moreover

$$[S_{i_r i_r}, S_{m, n}] = 0 \quad \text{if } \{i, j\} \cap \{m, n\} = \emptyset \quad (4.7)$$

also follows easily from the definition of  $S_{i_r i_r}$ . Suppose then that  $r < n$ . We shall show that  $[H_r, H_n] = 0$ . It suffices to show that  $[H_r, S_{i_n i_p}] = 0$  for  $r < n < p$ . But by (4.7)

$$[H_r, S_{i_n i_p}] = [S_{i_r i_r} + S_{i_r i_p}, S_{i_n i_p}]$$

which is zero by Lemma 4.2.

**COROLLARY 4.4.** *In the notation of Theorem 4.3 put*

$$E_j = - \sum_{i=1}^j H_i, \quad j = 1, \dots, 2k. \quad (4.8)$$

*Then the operators  $E_j$  commute with each other and*

$$H(x) = \sum_{j=1}^{2k-1} (y_{j+1} - y_j) E_j. \quad (4.9)$$

*Moreover*

$$E_j = -\left(\frac{1}{2}\right) \sum_{r, p=1}^j S_{i_r i_p} \quad (4.10)$$

and

$$E_j \geq 0, \quad j = 1, \dots, 2k \quad (4.11)$$

in the trace inner product on  $\mathcal{B}$ .

*Proof.* The operators  $E_j$  commute among themselves because the operators  $H_i$  commute, as shown in the proof of the theorem. Using (4.6) and a standard summation by parts identity we have  $H(x) = \sum_{j=1}^{2k-1} (y_{j+1} - y_j) E_j - y_{2k} E_{2k}$ . We show  $E_{2k} = 0$ . By (4.5) and (4.8) we have

$$\begin{aligned} E_{2k} &= - \sum_{r=1}^{2k} H_r \\ &= \left(\frac{1}{2}\right) \sum_{r=1}^{2k} S_{i_r, i_r} + \sum_{r=1}^{2k} \sum_{p>r} S_{i_r, i_p} \\ &= \left(\frac{1}{2}\right) \sum_{j=1}^{2k} S_{j,j} + \left(\frac{1}{2}\right) \sum_{i \neq j} S_{i,j} \\ &= \left(\frac{1}{2}\right) \sum_{i,j=1}^{2k} S_{i,j} \end{aligned}$$

which is zero on  $\mathcal{B}$  by Lemma 3.2. This proves (4.9). To prove (4.10) note that  $E_j = \sum_{r=1}^j (2^{-1} S_{i_r, i_r} + \sum_{p>r} S_{i_r, i_p}) = \sum_{r=1}^j (2^{-1} S_{i_r, i_r} - \sum_{p \leq r} S_{i_r, i_p})$  by Lemma 3.2. Remove the diagonal in the second sum and use the symmetry of  $S_{i,j}$  in  $i, j$  to get  $E_j = \sum_{r=1}^j -2^{-1} S_{i_r, i_r} - 2^{-1} \sum_{1 \leq r \neq p \leq j} S_{i_r, i_p} = -2^{-1} \sum_{r,p=1}^j S_{i_r, i_p}$ . To prove that  $E_j \geq 0$  put  $N_a^j B = -B T_a^{(j)}$  for  $1 \leq j \leq k$  and put  $N_a^j B = T_a^{(j-k)} B$  for  $k+1 \leq j \leq 2k$ . It follows from (3.7), (3.8), and (3.9) that  $S_{ij} = \sum_a N_a^i N_a^j$ . Moreover in the inner product  $(A, B) = \text{trace } B^* A$  each  $N_a^j$  is skew adjoint because each  $T_a^{(j)}$  is skew adjoint on  $(\mathbb{C}^N)^k$ . Further  $[N_a^j, N_a^i] = 0$  for all  $i, j$ . Hence if we put  $P_a = \sum_{r=1}^j N_a^{i_r}$  then (4.10) gives  $2E_j = -\sum_a P_a^2$  which is positive since  $P_a^* = -P_a$ .

*Remark 4.5.* If  $\{y_r\}_{r=1}^{2k}$  are all distinct then (4.10) can be written in the useful forms

$$\begin{aligned} -2E_j &= \sum_{r, p: y_r \leq y_j, y_p \leq y_j} S_{r,p} \\ &= \sum_{r, p: \sigma_r(x) \leq y_j, \sigma_p(x) \leq y_j} S_{r,p}. \end{aligned} \quad (4.12)$$

**COROLLARY 4.6.** If  $G$  is irreducible on  $\mathbb{C}^N$  then  $E_1 = E_{2k-1} = (\alpha/2)I$  where  $C = -\alpha I$  is the value of the Casimir operator.

*Proof.* Since  $S_{ii}$  is left or right multiplication by  $C^{(i)}$  we have

$$S_{ii} = -\alpha I, \quad i = 1, \dots, 2k. \quad (4.13)$$



Hence  $E_1 = -H_1 = \frac{1}{2}S_{i_1 i_1} + \sum_{p=2}^{2k} S_{i_1 i_p} = -\frac{1}{2}S_{i_1 i_1} + \sum_{j=1}^{2k} S_{i_1 j} = -\frac{1}{2}S_{i_1 i_1} = (\alpha/2)I$ .  
 Moreover  $E_{2k-1} = E_{2k} + H_{2k} = O - \frac{1}{2}S_{i_{2k} i_{2k}} = (\alpha/2)I$ .

**COROLLARY 4.7.** Assume that  $\sigma_1, \dots, \sigma_{2k}$  do not cross in  $[c, d]$  in the sense described in Theorem 4.3. Choose  $i_1, \dots, i_{2k}$  so that  $\sigma_{i_{j+1}}(x) \geq \sigma_{i_j}(x)$  for all  $x$  in  $[c, d]$  and  $j = 1, \dots, 2k-1$ . Let  $A_j(x) = \int_c^x (\sigma_{i_{j+1}}(t) - \sigma_{i_j}(t)) dt$ . Then the solution to Bralic's differential equation is

$$\phi(c, x) = \exp \left[ - \sum_{j=1}^{2k-1} A_j(x) E_j \right]. \quad (4.14)$$

*Proof.* Since  $H(x)$  is a commuting family of operators the solution to (3.13) with identity initial condition is  $\phi(c, x) = \exp[-\int_c^x H(x') dx]$  which, in view of (4.9) is (4.14).

In view of the appearance of the operators  $E_j$  in (4.14) as the coefficients of the areas between adjacent curves we shall refer to the  $E_j$  as the *area operators* associated to the curves  $\sigma_1, \dots, \sigma_{2k}$  over  $(c, d)$ . In particular  $E_j$  is the area operator associated to the region between  $\sigma_{i_j}$  and  $\sigma_{i_{j+1}}$  over  $(c, d)$ .

**COROLLARY 4.8.** Let  $\sigma_1, \dots, \sigma_{2k}$  be continuous real functions on  $[a, b]$  whose graphs have only finitely many intersection points with each other and denote by  $a_1 < a_2 < \dots < a_n$  the  $x$  coordinates in  $(a, b)$  of these intersection points. Denote by  $B$  the union of the graphs of  $\sigma_1, \dots, \sigma_{2k}$  over  $(a, b)$ . Let  $K_1, \dots, K_m$  denote the bounded connected components of  $((a, b) \times R) - B$ . Put  $a_0 = a$  and  $a_{n+1} = b$ . Over the interval  $(a_{r-1}, a_r)$  the curves  $\sigma_1, \dots, \sigma_{2k}$  do not cross and we denote by  $E^{r,p}$  the area operator on  $\mathcal{B}$  associated to  $K_p \cap ((a_{r-1}, a_r) \times R) \equiv K'_p$  if this intersection is nonempty. Otherwise we put  $E^{r,p} = 0$ . Then

(i)  $E^{r,p} = E^{r',p}$  if  $K'_p$  and  $K''_p$  are both nonempty. That is,  $E^{r,p}$  depends only on  $K_p$  wherever possible. The common value,  $E^p$ , will be called the area operator associated to  $K_p$ .

(ii)  $E^p$  and  $E^q$  commute if  $K_p$  and  $K_q$  have overlapping projections on the  $x$  axis. In particular if  $K_p$  projects onto all of  $(a, b)$  then  $E^p$  commutes with all  $E^q$ .

(iii) Bralic's map  $\phi(a, b)$  is a product of all the operators  $\{\exp[-|K_p| E^p]\}_{p=1}^m$  in some order where  $|K_p|$  denotes the area of  $K_p$ . If a boundary point of  $K_p$  lies on the left hand boundary of the strip, i.e., on the line  $x = a$ , then the factor  $\exp[-|K_p| E_p]$  may be placed on the far left in this product. If a boundary point of  $K_p$  lies on the right hand boundary of the strip then this factor may be placed on the far right in this product.

*Proof.* Each curve  $\sigma_j$  disconnects the strip  $(a, b) \times R$ . Hence two points in  $K_p$  both lie above  $\sigma_j$  or both lie below  $\sigma_j$ . Consequently if  $K'_p$  and  $K''_p$  are both nonempty then the same set of curves in  $\{\sigma_1, \dots, \sigma_{2k}\}$  lies below both. It follows from Remark 4.5 that  $E^{r,p} = E^{r',p}$ .

Now the projection of the connected open set  $K_p$  on the  $x$  axis is an open interval which is a union of one or more of the intervals  $(a_{r-1}, a_r)$  with some endpoints adjoined. If  $K_p$  and  $K_q$  have overlapping projections on the  $x$  axis then for some  $r$   $K_p^r$  and  $K_q^r$  are both non empty.  $E^{r,p}$  and  $E^{r,q}$  are two of the area operators over  $(a_{r-1}, a_r)$  discussed in Corollary 4.4 and therefore commute. This proves (ii).

To prove (iii), apply Corollary 4.7 to express the solution to Bralic's differential equation as a product of exponentials, one for each interval  $(a_{r-1}, a_r)$ . We have thus

$$\varphi(a, b) = \exp \left[ - \sum_{p=1}^m |K_p^1| E^{1,p} \right] \cdots \exp \left[ - \sum_{p=1}^m |K_p^{n+1}| E^{n+1,p} \right]$$

in that order (because  $H(x)$  appears on the right in (3.13)). By (i) we may write

$$\varphi(a, b) = \exp \left[ - \sum_{p=1}^m |K_p^1| E^p \right] \cdots \exp \left[ - \sum_{p=1}^m |K_p^{n+1}| E^p \right] \quad (4.15)$$

since the  $E^{r,p}$  that are defined as zero have  $|K_p^r| = 0$  anyway. Since  $|K_p| = \sum_{r=1}^{n+1} |K_p^r|$  it suffices to show that the factors in (4.15) can be rearranged suitably. Consider a connected component which, after relabeling, we may call  $K_1$  and which contains  $(a, a_1)$  in its projection on the  $x$  axis. If  $|K_1^r| \neq 0$  for exactly  $r = 1, \dots, j$  then  $E^{r,1} = E^1$  occurs with a non zero coefficient in each of the  $j$  leftmost factors in (4.15) and nowhere else. Moreover  $E^1$  commutes with all of the  $E^p$  in these  $j$  factors by (ii) and hence  $e^{-|K_1^r| E^1}$  may be commuted to the left from the  $r$ th factor for  $r = 1, \dots, j$  giving a factor  $\exp[-|K_1| E^1]$  on the far left and leaving no other factor involving  $E^1$  anywhere. Now we proceed by induction as follows. Having picked and relabeled components  $K_1, \dots, K_p$  and written  $\varphi(a, b) = \exp[-|K_1| E^1] \cdots \exp[-|K_p| E^p] \cdot F$  where  $F$  is a product of exponentials not involving  $E^1, \dots, E^p$  we choose from among the remaining connected components one whose projection on the  $x$ -axis extends at least as far to the left as any of the others that remain. Label this  $K_{p+1}$ . The argument given for  $K_1$  applies now to show that one can write  $F = \exp[-|K_{p+1}| E^{p+1}] F^1$  where  $F^1$  is a product of exponentials not involving  $E^{p+1}$ . This proves (iii) since a similar argument can be applied starting from the far right. We note that the  $K_1$  that we started with can be any component with a boundary point on the line  $x = a$ .

**COROLLARY 4.9.** *Let  $\{\sigma_n\}$  be a uniformly bounded sequence of continuous functions on  $[a, b]$ . Suppose that  $\sigma: [a, b] \rightarrow R$  is also continuous and that  $\sigma_n(x)$  converges to  $\sigma(x)$  for each  $x$  in  $(a, b)$ . Then  $U(\sigma_n)$  converges to  $U(\sigma)$  in  $L^p$  (white noise) for  $1 \leq p < \infty$ .*

*Proof.* Since the operators  $U(\sigma_n)$  are all unitary, hence, uniformly bounded, it suffices to prove convergence in  $L^2$ . But

$$E(\text{tr}[(U(\sigma_n) - U(\sigma))^* (U(\sigma_n) - U(\sigma))]) = 2 \text{Re tr } E[I - U(\sigma)^* U(\sigma_n)].$$

By (3.5)  $E(U(\sigma)^* U(\sigma_n)) = \varphi_n(a, b)I$  where  $\varphi_n(a, b)$  is the Bralic map for  $k=1$  determined by the two curves  $\sigma_n$  and  $\sigma$ . It is therefore the solution to the differential equation  $d\varphi_n(a, x)/dx = -\varphi_n(a, x)H_n(x)$  with initial condition  $\varphi_n(a, a) = I_{\mathcal{H}}$ . By (4.9)  $H_n(x)$  may be represented for each  $x$  in the form  $H_n(x) = \pm(\sigma_n(x) - \sigma(x))E_1$  where  $-2E_1 = S_{11}$  or  $S_{22}$ . Hence  $\|H_n(x)\|_{op} \leq \text{const.} |\sigma_n(x) - \sigma(x)|$ . Thus  $\|H_n(x)\|_{op}$  converges to zero boundedly on  $(a, b)$ . Hence  $\varphi_n(a, x)$  converges to its initial value  $I_{\mathcal{H}}$  for each  $x$  in  $[a, b]$  and in particular  $\varphi_n(a, b)I$  converges to  $I$ . Therefore  $\text{tr}[I - E(U(\sigma)^* U(\sigma_n))]$  converges to zero.

*Remark 4.10.* Suppose that  $\{\sigma_n\}$  is a sequence of continuous functions on  $[a, b]$  which converges pointwise and boundedly on  $[a, b]$  to a function  $\sigma$  which is itself continuous except at the points  $a_1 < a_2 < \dots < a_{q-1}$  in  $(a, b)$  and which has left and right hand limits at these points. Note that if  $\sigma'_n$  is the restriction of  $\sigma_n$  to  $(a_{r-1}, a_r)$  with  $a_0 = a$  and  $a_q = b$  then  $U(\sigma_n) = U(\sigma'_q) \cdots U(\sigma'_1)$ . By Corollary 4.9 each factor  $U(\sigma'_r)$  converges in  $L^p$  (white noise) for  $1 \leq p < \infty$  to  $U(\sigma^r)$ . It follows that  $U(\sigma_n)$  converges in  $L^p$  to  $U(\sigma^q) \cdots U(\sigma^1)$ . Since parallel translation along a vertical line segment is the identity operator this product may be interpreted as parallel translation along the curve formed from the graph of  $\sigma$  by joining the disconnected portions by vertical line segments above the jump points  $a_r$ . It would be natural therefore to define  $U(\sigma)$  by this product for such a piecewise continuous function  $\sigma$ . In effect we do this in the following sections. It is clear in fact from the definition (2.10) of parallel translation as well as the proof of Corollary 4.9 that  $U(\sigma)$  could be defined for any bounded measurable function  $\sigma$  on  $[a, b]$  and is continuous in  $\sigma$  in the  $L^1(a, b)$  norm into  $L^p$  (white noise) for  $1 \leq p < \infty$ . But the geometric meaning of  $U(\sigma)$  seems to be lost and we shall not use it.

## 5. CONNECTION OPERATORS

In order to evaluate the expectation of the product of the Wilson loop variables associated, for example, to two concentric circles it will be necessary to extend the previous formalism. Consider the configuration of curves shown in Figs. 5.1a and b. In both cases the number of curves lying above a point  $x$  on the  $x$  axis changes

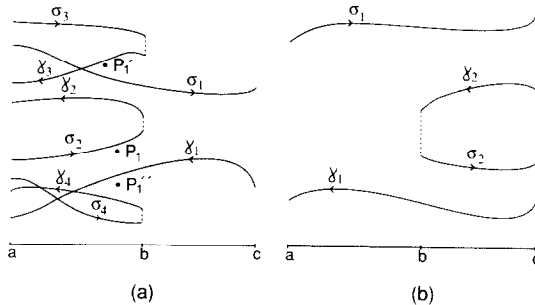


FIGURE 5.1

at  $b$  as  $x$  moves from left to right. Denote by  $\mathcal{B}_k$  the algebra of operators on  $(\mathbb{C}^N)^k$  which commute with  $g^{(k)}$  for all  $g$  in  $G$ . This was denoted by  $\mathcal{B}$  in Sections 3 and 4. But we want to use a different  $k$  over different intervals. Let  $m \leq k$ . We consider in general  $2m$  continuous real valued functions  $\sigma_1, \dots, \sigma_m, \gamma_1, \dots, \gamma_m$  on the bounded interval  $[a, c]$  and  $2(k-m)$  real valued continuous functions  $\sigma_{m+1}, \dots, \sigma_k, \gamma_{m+1}, \dots, \gamma_k$  on the subinterval  $[a, b]$ . Define  $J: \mathcal{B}_m \rightarrow \mathcal{B}_k$  by

$$JB = B \otimes I_{k-m}, \quad B \text{ in } \mathcal{B}_m, \quad (5.1)$$

where  $I_{k-m}$  is the identity operator on  $(\mathbb{C}^N)^{k-m}$ . Note that  $JB$  is indeed in  $\mathcal{B}_k$  since  $g^{(k)}(B \otimes I_{k-m})(g^{-1})^{(k)} = g^{(m)}B(g^{-1})^{(m)} \otimes I_{k-m} = B \otimes I_{k-m}$  for all  $g$  in  $G$ . We define

$$\varphi := \varphi(a, b) J\varphi(b, c): \mathcal{B}_m \rightarrow \mathcal{B}_k,$$

where  $\varphi(b, c): \mathcal{B}_m \rightarrow \mathcal{B}_m$  is defined as in Section 3 using the graphs of  $\sigma_1, \dots, \gamma_m$  over  $[b, c]$  and  $\varphi(a, b): \mathcal{B}_k \rightarrow \mathcal{B}_k$  is defined using the graphs of  $\sigma_1, \dots, \gamma_k$  over  $[a, b]$ . We assert that just as in Theorem 3.6 the map  $\varphi$  determines the expectation of products of all the Wilson loops obtainable from  $\sigma_1, \dots, \gamma_k$  as follows. We refer to the curves which are the graphs of  $\sigma_1, \dots, \sigma_m, \gamma_1, \dots, \gamma_m$  over  $[a, c]$  as the long curves and the graphs of  $\sigma_{m+1}, \dots, \sigma_k, \gamma_{m+1}, \dots, \gamma_k$  over the subinterval  $[a, b]$  as the short curves. Parallel translation along  $\sigma_j$  will always be taken from left to right and denoted  $U(\sigma_j)$  while parallel translation along  $\gamma_j$  will always be taken from right to left and denoted  $V(\gamma_j)$ . For  $j = m+1, \dots, k$  we join the short curve  $\sigma_j$  to the short curve  $\gamma_j$  by a vertical line segment above  $x = b$  from  $(b, \sigma_j(b))$  to  $(b, \gamma_j(b))$  as in Figure 5.1a. Above the endpoints  $a$  and  $c$  we join the curves by vertical line segments in a manner similar to that of Theorem 3.6. Specifically, we choose permutations  $P$  in  $S_m$  and  $Q$  in  $S_k$  to index the joining procedure. For  $j = 1, \dots, m$  we join  $\sigma_{P(j)}$  to  $\gamma_j$  by a vertical line segment above  $c$ . These two curves will then contribute a factor  $V(\gamma_j) U(\sigma_{P(j)})$  to our final Wilson loops. For  $j = 1, \dots, k$  we join  $\gamma_{Q(j)}$  to  $\sigma_j$  by a vertical line segment above  $a$ . These two curves will then contribute a factor  $U(\sigma_j) V(\gamma_{Q(j)})$  to our final Wilson loops. As in the discussion preceding Theorem 3.6 we obtain a finite number of closed curves  $\tau_j$ ,  $j = 1, \dots, r$ , for which

$$\begin{aligned} & \prod_{j=1}^r \text{trace } U(\tau_j) \\ &= \text{trace}[\hat{Q}(V(\gamma_1) \otimes \dots \otimes V(\gamma_k))(\hat{P} \otimes I_{k-m})(U(\sigma_1) \otimes \dots \otimes U(\sigma_k))]. \end{aligned} \quad (5.2)$$

Note that the factor  $\hat{P} \otimes I_{k-m}$  is the operator needed for connecting the long curves by  $P$  over  $c$  and connecting the short curves by the identity permutation over  $b$ .

LEMMA 5.1 (cf. Fig. 5.1a).

$$E \left[ \prod_{j=1}^r \text{trace } U(\tau_j) \right] = \text{trace}[\hat{Q}\varphi(a, b) J\varphi(b, c) \hat{P}]. \quad (5.3)$$

*Proof.* For  $j = 1, \dots, k$  write  $U_j$  for parallel translation along  $\sigma_j$  from  $a$  to  $b$  and  $V_j$  for parallel translation along  $\gamma_j$  from  $b$  to  $a$ . For  $j = 1, \dots, m$  write  $U'_j$  (respectively,  $V'_j$ ) for parallel translation along  $\sigma_j$  from  $b$  to  $c$  (respectively, along  $\gamma_j$  from  $c$  to  $b$ ). Then for the long curves  $U(\sigma_j) = U'_j U_j$  and  $V(\gamma_j) = V_j V'_j$ ,  $j = 1, \dots, m$  and for the short curves  $U(\sigma_j) = U_j$  and  $V(\tau_j) = V_j$ ,  $j = m + 1, \dots, k$ . All primed operators are independent of all unprimed operators. Substitute these products into the right side of (5.2) and take the expectation first with respect to the primed operators to get

$$\begin{aligned}
 & E \left[ \prod_{j=1}^r \text{trace } U(\tau_j) \right] \\
 &= E[\text{trace } \hat{Q}(V_1 \otimes \dots \otimes V_k) \\
 &\quad \times (E\{V'_1 \otimes \dots \otimes V'_m\} \hat{P}(U'_1 \otimes \dots \otimes U'_m) \otimes I_{k-m})(U_1 \otimes \dots \otimes U_k)] \\
 &= \text{trace } \hat{Q} E[(V_1 \otimes \dots \otimes V_k)(\{\varphi(b, c) \hat{P}\} \otimes I_{k-m}) U_1 \otimes \dots \otimes U_k] \\
 &= \text{trace } \hat{Q} \varphi(a, b) J \varphi(b, c) \hat{P} \quad (5.4)
 \end{aligned}$$

which proves (5.3).

The machinery for representing the situation of Fig. 5.1b is similar. The long curves are again the graphs of continuous functions  $\sigma_1, \dots, \sigma_m, \gamma_1, \dots, \gamma_m$  on the whole interval  $[a, c]$ , but this time short curves are the graphs of continuous functions  $\sigma_{m+1}, \dots, \sigma_k, \gamma_{m+1}, \dots, \gamma_k$  on the other subinterval  $[b, c]$ . Moreover the right hand connection permutation  $P$  is in  $S_k$  this time while  $Q$  is in  $S_m$ . We join the short curves by a vertical line segment over  $b$  which joins  $(b, \gamma_j(b))$  to  $(b, \sigma_j(b))$ ,  $j = m + 1, \dots, k$ . As before the configuration decomposes into closed curves  $\tau_1, \dots, \tau_r$  and we have this time

$$\begin{aligned}
 & \prod_{j=1}^r \text{trace } U(\tau_j) \\
 &= \text{trace}[(\hat{Q} \otimes I_{k-m})(V(\gamma_1) \otimes \dots \otimes V(\gamma_k)) \hat{P}(U(\sigma_1) \otimes \dots \otimes U(\sigma_k))] \quad (5.5)
 \end{aligned}$$

wherein the factor  $I_{k-m}$  effects the connections above  $b$ . As before we decompose the parallel translation operators along the long curves into independent factors and take first the expectation with respect to the operators over  $[b, c]$  to get

$$\begin{aligned}
 & E \left[ \prod_{j=1}^r \text{trace } U(\tau_j) \right] \\
 &= E[\text{trace}\{(\hat{Q} \otimes I_{k-m})(V_1 \otimes \dots \otimes V_m \otimes I_{k-m})(\varphi(b, c) \hat{P}) \\
 &\quad \times (U_1 \otimes \dots \otimes U_m \otimes I_{k-m})\}] \\
 &= E[\text{trace}\{\hat{Q}(V_1 \otimes \dots \otimes V_m)(J^* \varphi(b, c) \hat{P})(U_1 \otimes \dots \otimes U_m)\}],
 \end{aligned}$$

where  $J^*: \mathcal{B}_k \rightarrow \mathcal{B}_m$  is the adjoint of  $J$  relative to the trace inner products. Here  $J$

should be regarded as operating on all of  $\mathcal{B}((\mathbb{C}^N)^m)$  into  $\mathcal{B}((\mathbb{C}^N)^k)$  by  $JB = B \otimes I_{k-m}$ , but this does not affect the value of  $J^*$  on  $\mathcal{B}_k$  which is easily seen to lie in  $\mathcal{B}_m$ . Upon taking the final expectation we obtain the identity which reduces Fig. 5.1b to Bralic maps, namely

LEMMA 5.2 (cf. Fig. 5.1b).

$$E \left[ \prod_{j=1}^r \text{trace } U(\tau_j) \right] = \text{trace} [\hat{Q}\varphi(a, b) J^* \varphi(b, c) \hat{P}]. \quad (5.6)$$

We shall refer to the operators  $J$  and  $J^*$  as *connection operators* since they contain the algebraic information which effects the connection of the short curves along vertical line segments at  $x = b$  in Figs. 5.1a and 5.1b.

In addition to these connection operators we shall need another type of connection operator. Suppose that in Fig. 5.1a the curves  $\sigma_2$  and  $\gamma_2$  continue on to the left of  $a$  over some interval  $[d, a]$  while  $\sigma_1$  and  $\gamma_1$  stop at  $a$  as shown. Over the interval  $[d, a]$  we have  $k = 1$ . The subscripts that label our curves are important because they determine the position of the corresponding parallel translation operators  $U_j$  and  $V_j$  in Eq. (3.5). Thus the curves  $\sigma_2$  and  $\gamma_2$  should be labeled  $\sigma_1$  and  $\gamma_1$  over  $[d, a]$  but cannot be so labeled over  $[a, b]$  without causing labeling inconsistency for the other two curves over  $[a, c]$  which are already shown in Fig. 5.1a.

We define therefore a relabeling operator as follows. With  $a < b < c$  as before we suppose that we are given  $2k$  continuous functions whose restrictions to  $[b, c]$  we label  $\sigma'_1, \dots, \sigma'_k, \gamma'_1, \dots, \gamma'_k$  and whose corresponding restrictions to  $[a, b]$  we label  $\sigma_{i_1}, \dots, \sigma_{i_k}, \gamma_{j_1}, \dots, \gamma_{j_k}$ . Thus  $\sigma_{i_r}$  and  $\sigma'_r$  are restrictions of the same continuous function to adjacent intervals. Write  $\alpha(n) = j_n$  and  $\beta(r) = i_r$ . Then  $\alpha$  and  $\beta$  are in  $S_k$ . Define

$$RB = \hat{\alpha}^{-1} B \hat{\beta} \quad (5.7)$$

for  $B$  in  $\mathcal{B}_k$ . Clearly  $RB$  is also in  $\mathcal{B}_k$ .

LEMMA 5.3. *The map  $\varphi := \varphi(a, b) R\varphi(b, c)$  determines the expectation of Wilson loops in the canonical way:  $\text{tr}(\hat{Q}\varphi\hat{P})$  is the expectation of the product of Wilson loops determined by connecting the continuous curves at  $a$  and  $c$  by  $Q$  and  $P$ , respectively.*

*Proof.* Write  $U_j = U(\sigma_j)$ ,  $V_j = V(\gamma_j)$ ,  $U'_j = U(\sigma'_j)$ , and  $V'_j = V(\gamma'_j)$  and observe the identity

$$\begin{aligned} \text{tr}[\hat{Q}(V_1 \otimes \dots \otimes V_k) \hat{\alpha}^{-1} \{(V'_1 \otimes \dots \otimes V'_k) \hat{P}(U'_1 \otimes \dots \otimes U'_k)\} \hat{\beta}(U_1 \otimes \dots \otimes U_k)] \\ = \text{tr}[\hat{\beta} \hat{Q} \hat{\alpha}^{-1} ((V_{j_1} V'_1) \otimes \dots \otimes (V_{j_k} V'_k)) \hat{P}((U'_1 U_{i_1}) \otimes \dots \otimes (U'_k U_{i_k}))]. \end{aligned} \quad (5.8)$$

By independence of primed from unprimed operators the expectation of the left side is  $\text{trace}(\hat{Q}\varphi\hat{P})$ . But by Corollary 3.9 the right side is a product of traces of disjoint operators loops in which the typical sequences that appear in a loop are  $V_{j_r} V'_r U'_{P(r)} U_{i_{P(r)}}$  and  $U'_r U_{i_r} V_{j_s} V'_s$  where  $j_s = j_{\alpha^{-1}Q\beta(r)} = \alpha\alpha^{-1}Q\beta(r) = Qi_r$ . Thus  $V_{j_r}$  is

preceded by  $V'_r$  as desired,  $V'_r$  is preceded by  $U'_{p(r)}$  as desired, and  $U_i$  is preceded by  $V_{Q_r}$  as desired.

**EXAMPLE 5.4.** Let  $C_1$  and  $C_2$  be two intersecting circles with centers on the  $x$  axis and oriented clockwise. Denote by  $a < b < c < d$  the intersection points with the  $x$  axis. Let  $\sigma_1$  and  $\gamma_1$  be the functions on  $[a, c]$  whose graphs give the upper and lower halves of the leftmost circle  $C_1$  while  $\sigma_2$  and  $\gamma_2$  are the functions on  $[b, d]$  whose graphs give the upper and lower halves of  $C_2$ . Let  $\alpha = \beta = \tau$  be the transposition in  $S_2$  and denote by  $R: \mathcal{B}_2 \rightarrow \mathcal{B}_2$  the relabeling operator;  $RB = \hat{\tau}^{-1}B\hat{\tau}$ . Write  $JB = B \otimes I_1$  for  $B$  in  $\mathcal{B}_1$  and  $I_1 =$  identity operator on  $\mathbb{C}^N$  as in (5.1) with  $k = 2$ ,  $m = 1$ . Pick a point  $c'$  with  $b < c' < c$ . Then the Bralic map for the two circles is

$$\varphi = \varphi(a, b) J^* \varphi(b, c') R \varphi(c', c) J \varphi(c, d),$$

where  $\varphi(a, b)$  is determined by  $\sigma_1$  and  $\gamma_1$  over  $[a, b]$ ,  $\varphi(b, c')$  is determined by  $\sigma_1, \sigma_2, \gamma_1, \gamma_2$  (in this order) over  $[b, c']$ ,  $\varphi(c', c)$  is determined by  $\sigma_2, \sigma_1, \gamma_2, \gamma_1$  (in this order) over  $[c', c]$ , and  $\varphi(c, d)$  is determined by  $\sigma_2, \gamma_2$  over  $[c, d]$ . The right hand factor  $J$  closes the circle  $C_1$  on the right (as in Figure 5.1a) by a vertical segment (of length zero) and at the same time connects  $\sigma_2 | [c', c]$  to  $\sigma_2 | [c, d]$ . It also connects  $\gamma_2 | [c', c]$  to  $\gamma_2 | [c, d]$ . The relabeling operator  $R$  reverses the order of factors at  $c'$  so that  $J^*$  may close the circle  $C_2$  at  $b$  and properly connect  $\sigma_1 | [a, b]$  to  $\sigma_1 | [b, c']$  and  $\gamma_1 | [a, b]$  to  $\gamma_1 | [b, c']$ . We may simplify  $\varphi$  by observing that  $\varphi(c', c)$  converges to  $I_{\mathcal{B}_2}$  as  $c' \nearrow c$  while  $\varphi(b, c')$  converges to  $\varphi(b, c)$ . Hence we have  $\varphi = \varphi(a, b) J^* \varphi(b, c) R J \varphi(c, d)$ . Here  $RJ$  closes the circle  $C_1$  at  $c$  and simultaneously makes the correct lateral linking of  $\sigma_2$  on the right and left of  $c$ . Alternatively we could have put  $c' = b$  obtaining  $\varphi = \varphi(a, b) J^* R \varphi(b, c) J \varphi(c, d)$  in which  $\varphi(b, c)$  now has a different meaning since it is determined by  $\sigma_2, \sigma_1, \gamma_2, \gamma_1$  instead of  $\sigma_1, \sigma_2, \gamma_1, \gamma_2$ . To get the expectation of the product of the two Wilson loops take  $P = Q = I$  (which is the only choice since  $k = 1$  on the end intervals). The expectation is then  $\text{trace}[\varphi I_1]$ . It follows from Corollary 4.8 and the equation  $\varphi = \varphi(a, b) J^* \varphi(b, c) R J \varphi(c, d)$  that the expectation is a function of the seven finite areas determined by the two circles and the two inner vertical tangent lines. But in the next section we shall show that it depends only on the areas of the three bounded connected components determined by the two circles.

**Remark 5.5.** If two or more curves should close on the left over some point  $b$  (cf. Fig. 5.1b) while two or more other curves close on the right over the same point  $b$  (cf. Fig. 1a) then the overall Bralic map for this configuration can be constructed as before in either of the forms  $J_1 J_2^*$  or  $J_3^* J_4$  where the  $J_i$  are injections as in (5.1). This can be derived directly as in the previous cases or by simply applying the results for Figs. 5.1a and 5.1b consecutively in either order.

**Remark 5.6.** As noted in Remark 4.10 parallel translation along the graphs of piecewise continuous functions is a well defined notion. Moreover one need only look at Eqs. (5.4), (5.5), and (5.8) to see that the linking operators  $J$ ,  $J^*$ , and  $R$

over a point  $b$  implement the lateral connection of two graphs even if there is a jump discontinuity of the combined graphs at  $b$ .

We summarize the results of this section in the following theorem.

**THEOREM 5.7.** *Suppose that  $a_0 < a_1 < \dots < a_n$  are real numbers and  $k_1, \dots, k_n$  are strictly positive integers. Let  $\sigma'_1, \dots, \sigma'_{k_r}, \gamma'_1, \dots, \gamma'_{k_r}$  be continuous real functions on  $[a_{r-1}, a_r]$ . There exist operators  $\hat{J}_i: \mathcal{B}_{k_{i+1}} \rightarrow \mathcal{B}_{k_i}$ ,  $i = 1, \dots, n-1$  such that the operator  $\varphi: \mathcal{B}_{k_n} \rightarrow \mathcal{B}_{k_1}$  defined by*

$$\varphi = \varphi(a_0, a_1) \hat{J}_1 \varphi(a_1, a_2) \hat{J}_2 \cdots \hat{J}_{n-1} \varphi(a_{n-1}, a_n)$$

*determines the expectation of all Wilson loops constructed from the curves  $\sigma'_i, \gamma'_i$  by specified interior lateral and vertical connections at  $a_1, \dots, a_{n-1}$  as in the preceding cases. For vertical connections at the endpoints  $a_0$  and  $a_n$  specified by permutations  $Q$  in  $S_{k_1}$  and  $P$  in  $S_{k_n}$  the expectation is  $\text{trace}(\hat{Q}\varphi\hat{P})$ . Each  $\hat{J}_i$  is a product of an injection  $J$  as in (5.1), the adjoint of such an injection, and a relabeling operator  $R$  such as in (5.7). One or more of these three factors could be the identity operator.*

**Remark 5.8.** The vertical connections at the endpoints  $a_0$  and  $a_n$  can be represented by connection operators also. Define  $\mathcal{B}_0$  to be  $\mathbb{C}$  and extend the definition (5.1) to  $m = 0$  by putting  $J^{(k)}z = zI_k$ , for  $z$  in  $\mathbb{C}$ . Choose  $\alpha = I$  and  $\beta = P$  in (5.7) to get the relabeling operator  $R_P B = B\hat{P}$ . Choose  $\alpha^{-1} = Q$  and  $\beta = I$  to get the relabeling operator  $L_Q B = \hat{Q}B$ . Then, in the notation of Theorem 5.7 we see that  $J^{(k_1)*} L_Q \varphi R_P J^{(k_n)}$  is an operator on  $\mathbb{C}$  consisting of multiplication by the number  $(J^{(k_1)*} L_Q \varphi R_P J^{(k_n)} 1, 1) = (\hat{Q}\varphi\hat{P}, J^{(k_1)} 1) = (\hat{Q}\varphi\hat{P}, I_{k_1}) = \text{trace}(\hat{Q}\varphi\hat{P})$ , which is the expectation of the desired products of Wilson loops.

## 6. A STANDARD FORM FOR BRALIC'S MAP

Let  $\sigma_1, \dots, \sigma_{k_1}, \gamma_1, \dots, \gamma_{k_1}$  be continuous functions on  $[a, b]$  while  $\sigma'_1, \dots, \sigma'_{k_2}, \gamma'_1, \dots, \gamma'_{k_2}$  are continuous functions on  $[b, c]$ . We suppose specified lateral and vertical connections among the graphs of these functions at  $x = b$  as in Section 5, so that some curves extend continuously across  $b$  while others are joined by vertical line segments at  $b$  to curves on the same side of  $b$  but going in the opposite direction (i.e., a  $\sigma_i$  to a  $\gamma_j$ ). The linking operator at  $b$  that accomplishes this may be chosen of the form  $\hat{J} = R_1 J_1 R_2 J_2^* R_3$  where the  $R_i$  are relabeling operators. Let  $W$  be the union of the graphs of all these functions,  $\sigma_i, \gamma_j, \sigma'_i, \gamma'_j$ , along with the vertical segments. If  $K_1$  is a connected component of  $((a, b) \times R) - W$  and  $K_2$  is a connected component of  $((b, c) \times R) - W$  we shall say that  $K_1$  and  $K_2$  are adjacent (at  $b$ ) if there is a curve in  $((a, c) \times R) - W$  joining some point  $p_1$  in  $K_1$  to some point  $p_2$  in  $K_2$ . In Fig. 6.1,  $K_1, K_2$  is an adjacent pair.  $K_1, K'_2$  is another adjacent pair.



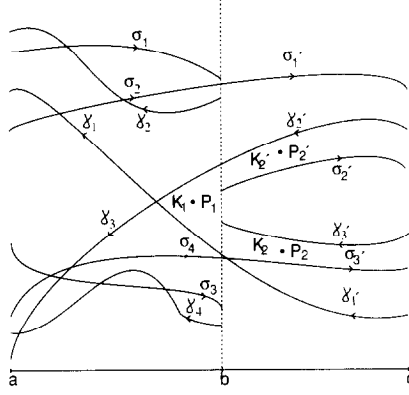


FIGURE 6.1

**THEOREM 6.1.** Suppose that  $K_1$  and  $K_2$  are adjacent components at  $b$  and that  $E: \mathcal{B}_{k_1} \rightarrow \mathcal{B}_{k_1}$  and  $F: \mathcal{B}_{k_2} \rightarrow \mathcal{B}_{k_2}$  are the associated area operators. Then

$$E\hat{J} = \hat{J}F. \quad (6.1)$$

The proof relies on the following lemmas. We shall use this theorem in Section 6 and Section 7 to show that the expectation of products of Wilson loops depends in a certain sense on the areas of the bounded connected components of the complement of the loops.

**LEMMA 6.2.** Suppose that  $\hat{J}$  is simply an injection  $J$  as in Eq. (5.1) and  $K_1$  and  $K_2$  are bounded. Then Theorem 6.1 holds.

*Proof.* This is the situation of Fig. 5.1a which also illustrates a pair  $p_1, p_2$ . Write  $k_1 = k$  and  $k_2 = m < k$  for this case and put  $p_i = (x_i, z_i)$  for  $i = 1, 2$ . Using (4.9) we have

$$H(x_1) = \sum_{j=1}^{2k-1} (y_{j+1} - y_j) E_j$$

and

$$H(x_2) = \sum_{n=1}^{2m-1} (y'_{n+1} - y'_n) E'_n,$$

where  $E_j$  acts on  $\mathcal{B}_k$ ,  $E'_n$  acts on  $\mathcal{B}_m$ , the numbers  $y_j$  are the ordinates  $\{\sigma_i(x_1), \gamma_i(x_1)\}_{i=1}^k$  labeled in some nondecreasing order, and  $y'_n$  are obtained the same way from  $\{\sigma_i(x_2), \gamma_i(x_2)\}_{i=1}^m$ . Since  $p_1$  and  $p_2$  do not lie on any graph there is a unique  $j$  and a unique  $n$  such that

$$y_j < z_1 < y_{j+1} \quad \text{and} \quad y'_n < z_2 < y'_{n+1}. \quad (6.2)$$

Then the area operator for  $K_1$  is  $E_j$  and the area operator for  $K_2$  is  $E'_n$ . We must show that

$$E_j J = J E'_n. \quad (6.3)$$

As in Section 5, for  $i = 1, \dots, m$ , we refer to  $\sigma_i \vee \sigma'_i$  and  $\gamma_i \vee \gamma'_i$  as the long curves. The remaining graphs, over  $[a, b]$ , will be called the short curves. Since any long curve disconnects the vertical strip  $(a, c) \times R$  each long curve lies above both points  $p_1$  and  $p_2$  or lies below both. Hence the set of long curves lying below  $p_1$  is the same as the set lying below  $p_2$ . By (4.10) we have

$$-2E_j = \sum_{r, p=1}^j S_{i_r, i_p}. \quad (6.4)$$

Here  $S_{i_r, i_p}$  acts on  $\mathcal{B}_k$  and  $p$  and  $r$  run over a subset of  $\{1, \dots, 2k\}$ . Let  $L = \{r : i_r \in \{1, \dots, m, k+1, \dots, k+m\}\}$ . Then  $L$  consists of the indices  $r$  for which  $i_r$  refers to a long curve in the notation  $\sigma_{k+r} = \gamma_r$ . We write  $\sum'_r$  for the sum over  $\{1, \dots, j\} \cap L$  and  $\sum''_r$  for the sum over  $\{1, \dots, j\} - L$ . Define

$$-2F_j = \sum'_r \sum'_p S_{i_r, i_p}.$$

It follows from (6.4), from the similar definition of  $E'_n$ , from the definition of  $S_{i_r, i_p}$ , and from the fact that the set of long curves lying below  $p_1$  is the same set as that for  $p_2$ , that

$$F_j(B \otimes I_{k-m}) = (E'_n B) \otimes I_{k-m} \quad \text{for } B \text{ in } \mathcal{B}_m.$$

It suffices therefore to show that  $(E_j - F_j)(B \otimes I_{k-m}) = 0$ . Now if  $m < i \leq k$  then for any  $p$  in  $\{1, \dots, 2k\}$  we have

$$(S_{i, p} + S_{k+i, p})(B \otimes I_{k-m}) = 0 \quad (6.5)$$

because  $T_a^{(i)}$  commutes with  $B \otimes I_{k-m}$  so that the left side of (6.5) is (e.g., for  $1 \leq p \leq k$ )  $\sum_a [(B \otimes I_{k-m}) T_a^{(p)} T_a^{(i)} - T_a^{(i)} (B \otimes I_{k-m}) T_a^{(p)}] = 0$ . If  $m < i \leq k$  then  $p_1$  cannot lie between the graphs of the short curves  $\sigma_i$  and  $\gamma_i$  because the open set lying between these graphs is disconnected from the rest of  $(a, c) \times R$  by  $W$ . Hence  $p_1$  lies above both graphs or neither. It therefore follows from (6.5) that  $\sum''_r S_{i_r, i_p}(B \otimes I_{k-m}) = 0$  since  $\sum''_r$  is a sum over all short curves lying below  $p_1$ . Hence

$$-2(E_j - F_j)(B \otimes I_{k-m}) = \left( \sum'_p \sum''_r + \sum''_p \sum'_r + \sum''_p \sum''_r \right) S_{i_r, i_p}(B \otimes I_{k-m}) = 0.$$

LEMMA 6.3. Suppose that  $\hat{J} = J$  as in Lemma 6.1 but that  $K_2$  is unbounded. Then

$$EJ = 0. \quad (6.6)$$

*Proof.* We continue the notation of the proof of Lemma 6.2. We must show that  $E_j J = 0$ . Since  $p_1$  is in an unbounded component of  $((a, c) \times R) - W$  either there are no long curves which lie below  $p_1$  (as, e.g.,  $p'_1$  in Fig. 5.1a) or all long curves lie below  $p_1$  (as, e.g.,  $p''_1$ ). In the former case we have in the notation of Lemma 6.2,  $-2E_j = \sum_r'' \sum_p'' S_{i_r, i_p}$  and since  $p_1$  lies above both short curves  $\sigma_i$  and  $\gamma_i$  or above neither we have, as before, that  $\sum_r'' S_{i_r, i_p}(B \otimes I_{k-m}) = 0$ . Hence  $E_j J = 0$ . In the latter case  $p_1$  lies above all curves with the exception of an even number of short curves which come in matched pairs  $\sigma_i, \gamma_i$ . If we denote by  $\sum_r'''$  a sum over these matched pairs then  $\sum_r''' S_{i_r, i_p}(B \otimes I_{k-m}) = 0$  because of (6.5). Hence

$$\begin{aligned} -2E_j(B \otimes I_{k-m}) &= \sum_{p=1}^j \sum_{r=1}^j S_{i_r, i_p}(B \otimes I_{k-m}) + \sum_{p=1}^j \sum_r''' S_{i_r, i_p}(B \otimes I_{k-m}) \\ &= \sum_{p=1}^j \sum_{i=1}^{2k} S_{i, i_p}(B \otimes I_{k-m}) = 0 \quad \text{by (3.10).} \end{aligned}$$

*Remark 6.4.* By (4.9)  $E_{2k} = 0$  and hence the area operator of an unbounded component is zero. Thus by Lemma 6.3, Theorem 6.1 holds if  $\hat{J} = J$  and  $K_2$  is unbounded. It clearly holds if both  $K_1$  and  $K_2$  are unbounded.

**LEMMA 6.5.** *Suppose that  $\hat{J}$  is the adjoint  $J^*$  of an injection. Then Theorem 6.1 holds. In particular if  $K_1$  is unbounded then  $J^* F = 0$ .*

*Proof.* To take advantage of the extensive notation developed in Lemma 6.2 we interchange the left and right intervals in this proof. Thus we take the unprimed functions  $\sigma_i, \gamma_i, i = 1, \dots, k$  to be defined over  $[b, c]$  while  $\sigma'_j, \gamma'_j, j = 1, \dots, m$  are defined over  $[a, b]$ . Moreover  $\sigma_i$  is connected to  $\gamma_i$  at  $b$  for  $i = m+1, \dots, k$ . The short curves therefore lie over  $[b, c]$  now. This is the situation for which the linking operator is  $J^*$  where  $J: \mathcal{B}_m \rightarrow \mathcal{B}_k$  is given again by (5.1) as in Lemma 6.2. Assume  $p_1$  lies over  $(b, c)$  now while  $p_2$  lies over  $(a, b)$ . We let  $E_j: \mathcal{B}_k \rightarrow \mathcal{B}_k$  be the area operator associated to the component  $K_2$  of  $((b, c) \times R) - W$  containing  $p_1$  and let  $E'_n: \mathcal{B}_m \rightarrow \mathcal{B}_m$  be that associated to the component  $K_1$  of  $((a, b) \times R) - W$  containing  $p_2$ . As before  $p_1$  and  $p_2$  are assumed to lie in the same connected component of  $((a, c) \times R) - W$ . Then we must show

$$J^* E_j = E'_n J^* \quad (6.7)$$

if  $K_1$  is bounded and

$$J^* E_j = 0, \quad (6.8)$$

if  $K_1$  is unbounded. The algebraic definitions of  $J, E_j$ , and  $E'_n$  are the same as in Lemma 6.2 even though the associated picture is different (the short curves now lie over  $[b, c]$ ). Hence by Lemma 6.2,  $E_j J = J E'_n$ . By Corollary 4.4  $E_j$  and  $E'_n$  are Hermitian in the trace inner products on  $\mathcal{B}_k$  and  $\mathcal{B}_m$ , respectively. Hence the adjoint of (6.3) gives (6.7) while the adjoint of (6.6) gives (6.8).

LEMMA 6.6. *If  $\hat{J}$  is simply a relabeling operator  $R$  then Theorem 6.1 holds.*

*Proof.* In this case the number,  $k$ , of curves over  $[a, b]$  is the same as the number over  $[b, c]$ . We use the notation of Lemma 5.3.  $p_1$  lies over the interval  $(a, b)$  on the  $x$ -axis and  $p_2$  lies over  $(b, c)$ .  $E_j$  is the area operator for  $K_1$  and  $E'_n$  is the area operator for  $K_2$ . Both act on  $\mathcal{B}_k$  as in Lemma 6.2. We wish to show that  $E_j R = R E'_n$ .

We may assume that the connected components in question are bounded since otherwise  $E_j = E'_n = 0$ . Let  $\psi$  be the permutation in  $S_{2k}$  given by  $\psi(i) = \beta(i)$  and  $\psi(k+i) = k + \alpha(i)$  for  $i = 1, \dots, k$ . Using the definitions (3.6) in Section 3 it is straightforward to verify that  $\hat{\alpha}^{-1} T_a^{(i)} \hat{\alpha} = T_a^{(\alpha(i))}$  and hence  $R T_a^{(i)} (R^{-1} B) = T_a^{(\alpha(i))} B$  while  $R \{ (R^{-1} B) T_a^{(i)} \} = B T_a^{(\beta(i))}$  for  $i = 1, \dots, k$  where  $RB = \hat{\alpha}^{-1} B \hat{\beta}$  for all  $B$  in  $\text{End}(\mathbb{C}^N)^k$ . Consequently  $R S_{ij} R^{-1} = S_{\psi(i), \psi(j)}$  for  $i, j = 1, \dots, 2k$  in view of the definitions (3.7), (3.8), and (3.9). We write  $\sigma_i(x) = \gamma_{i-k}(x)$  and  $\sigma'_i = \gamma'_{i-k}$  for  $i = k+1, \dots, 2k$  in order to apply the notation of Section 4. By (4.10) we have  $-2E_j = \sum_{r, p \in A} S_{r, p}$  and  $-2E'_n = \sum_{r, p \in B} S_{r, p}$  where  $A = \{s \in \{1, \dots, 2k\} : \sigma_s(x_1) < z_1\}$  and  $B = \{r \in \{1, \dots, 2k\} : \sigma'_r(x_2) < z_2\}$ . But since  $p_1$  and  $p_2$  lie in the same connected component of  $((a, c) \times R) - W$  we have  $\sigma'_r(x_2) < z_2$  if and only if  $\sigma_{\psi(r)}(x_1) < z_1$ . Hence  $A = \psi(B)$ . Thus  $-2R E'_n R^{-1} = \sum_{r, p \in B} S_{\psi(r), \psi(p)} = \sum_{s, q \in A} S_{s, q} = -2E_j$ .

*Proof of Theorem 6.1.* By the previous lemmas the theorem holds in case  $\hat{J}$  is any one of the factors  $J_1$ ,  $J_2^*$ , or  $R_i$ . We get all the other cases by a procedure which we illustrate for the case  $\hat{J} = J_1 J_2^*$ . Pick a curve  $C$  in  $((a, c) \times R) - W$  with endpoints  $p_1$  and  $p_2$  in  $K_1$  and  $K_2$ , respectively. We may assume  $p_1$  lies to the left of  $x = b$ . By assumption  $K_1$  lies in  $(a, b) \times R$ . So we may choose  $b' < b$  so that  $p_1$  is to the left of the line  $x = b'$ . For each pair  $\sigma_i, \gamma_i$  of curves over  $[a, b]$  which are connected to each other at  $b$  by  $J_1$  let us shift the point of connection to  $b'$ , removing from  $W$  the vertical line segment at  $x = b$  which makes the connection and replacing it by the vertical line segment from  $(b', \sigma_i(b'))$  to  $(b', \gamma_i(b'))$  which makes the new connection. Furthermore remove the leftover portions of the graphs of  $\sigma_i$  and  $\gamma_i$  which lie over  $(b', b]$ .  $W$  has thereby been replaced by a new closed set  $W'$ . Let  $K_3$  be the connected component of  $((b', b) \times R) - W'$  through which the curve  $C$  passes and let  $G: \mathcal{B}_m \rightarrow \mathcal{B}_m$  be the associated area operator. Then  $G J_2^* = J_2^* F$  by Lemma 6.5 and  $E J_1 = J_1 G$  by Lemmas 6.2 and 6.3. Combining these gives  $E J_1 J_2^* = J_1 J_2^* F$ .

REMARK 6.7. Although it was not necessary to take the limit  $b' \nearrow b$  in the last proof the convergence is easily established because parallel translation  $U(x)$  along the graph of a continuous function  $\sigma$  from  $c$  to  $x$  is continuous into  $L^p$  (white noise) for all  $p < \infty$  as a function of  $x$  since it is the solution of a stochastic differential equation and continuity into  $L^2$  for unitaries  $U(x)$  implies continuity into  $L^p$ .

THEOREM 6.8. *Let  $a = a_0 < a_1 < \dots < a_n = b$  be real numbers. Suppose that  $\sigma_1^r, \dots, \sigma_{k_r}^r, \gamma_1^r, \dots, \gamma_{k_r}^r$  are continuous real valued functions on  $[a_{r-1}, a_r]$  for  $r = 1, \dots, n$ .*

For  $r = 1, \dots, n-1$  we assume that some of the curves  $\sigma_j^r$  are connected on the right by vertical line segments above  $a_r$  to corresponding curves  $\gamma_i^r$  in the manner described in Section 5 while some of the curves  $\sigma_i^{r+1}$  are connected on the left above  $a_r$  to some corresponding  $\gamma_j^{r+1}$ . The other curves are connected laterally to curves over the adjacent intervals, cf. Fig. 6.1. All of this information is encoded in a connection operator  $\hat{J}_r: \mathcal{B}_{k_r} \rightarrow \mathcal{B}_{k_{r-1}}$  for each  $r = 1, \dots, n-1$  as described in Section 5. Let  $\varphi_r = \varphi(a_{r-1}, a_r): \mathcal{B}_{k_r} \rightarrow \mathcal{B}_{k_r}$  be the Bralic map for  $[a_{r-1}, a_r]$  and write  $\varphi = \varphi_1 \hat{J}_1 \varphi_2 \hat{J}_2 \cdots \varphi_n$  for the overall Bralic map for  $[a, b]$ .

Denote by  $W$  the union of the graphs of all the functions  $\sigma_i^r, \gamma_i^r$  together with the vertical connecting line segments above the points  $a_1, \dots, a_{n-1}$ . We assume that the graphs intersect in only finitely many points and that the vertical line segments are disjoint. Denote by  $K_1, \dots, K_p$  the bounded connected components of  $((a, b) \times R) - W$ . For each component  $K_s$  pick a vertical strip  $(a_{r-1}, a_r) \times R$  which intersects  $K_s$  and let  $E^{(s)}$  denote the area operator on  $\mathcal{B}_{k_r}$  associated to any one of the nonempty connected components  $K_s^0$  of  $K_s \cap ((a_{r-1}, a_r) \times R)$ . Let  $A_s$  be the area of  $K_s$ . Then  $\varphi$  is a product in some order of the linking operators  $\hat{J}_1, \dots, \hat{J}_{n-1}$  and the operators  $\exp[-A_s E^{(s)}]$ ,  $s = 1, \dots, p$ .

*Proof.* The projection of a component  $K_s$  onto the  $x$ -axis is an interval which intersects one or more adjacent intervals  $(a_{j-1}, a_j)$ . For such a  $j$ ,  $K_s \cap ((a_{j-1}, a_j) \times R)$  is a nonempty open set. Let  $K'$  be one of its components and let  $E: \mathcal{B}_{k_j} \rightarrow \mathcal{B}_{k_j}$  be the area operator associated to  $K'$  by Corollary 4.8.  $\varphi_j$  is a product of exponentials one of which is  $\exp[-|K'| E]$ . Moreover if  $K'$  borders the left (respectively, right) edge of the strip  $(a_{j-1}, a_j) \times R$  then this exponential factor may be placed on the left (respectively, right) side of the product by Corollary 4.8. Pick a point in  $K_s^0$  and a point in  $K'$  and choose a piecewise linear curve in  $K_s$  which connects these two points, which has no vertical segments and has no changes of slope above the points  $a_0, \dots, a_n$ . This is possible because  $K_s$  is open and connected. If  $K' \neq K_s^0$  then the furthest point along the curve which is not in  $K'$  lies above  $a_{j-1}$  or  $a_j$  and therefore the curve enters  $K'$  from the left or right boundary of the strip, respectively. In the former case  $K'$  borders the left edge of the strip  $(a_{j-1}, a_j) \times R$  and the curve enters  $K'$  from a connected component  $K''$  of  $K_s \cap ((a_{j-2}, a_{j-1}) \times R)$  whose area operator  $F: \mathcal{B}_{k_{j-1}} \rightarrow \mathcal{B}_{k_{j-1}}$  satisfies  $F \hat{J}_{j-1} = \hat{J}_{j-1} E$  by Theorem 6.1. Hence  $\exp[-|K'| F] \hat{J}_{j-1} = \hat{J}_{j-1} \exp[-|K'| E]$ . Thus the factor  $\exp[-|K'| E]$ , which can be placed on the left of  $\varphi_j$ , can be commuted to the left of  $\hat{J}_{j-1}$  if one replaces  $E$  by  $F$ . If the curve enters  $K'$  from the right edge of the strip  $(a_{j-1}, a_j) \times R$  we can similarly commute this factor in  $\varphi_j$  past  $\hat{J}_j$  upon replacing  $E$  by some area operator on  $\mathcal{B}_{k_{j+1}}$ . By following backwards along the piecewise linear curve to  $K_s^0$  one may, in this manner, commute the factor  $e^{-|K'| E}$  in  $\varphi$  past the connection operators and Bralic maps  $\varphi_i$  by Theorem 6.1 and Corollary 4.8, changing  $E$  when necessary, till one arrives at  $K_s^0$ . The factor  $e^{-|K'| E}$  has then changed to  $e^{-|K'| E^{(s)}}$  which may be combined with the factor  $e^{-|K_s^0| E^{(s)}}$  in  $\varphi$ , to give a factor  $\exp[-(|K'| + |K_s^0|) E^{(s)}]$ . If we carry out this procedure for each connected component  $K'$  of every nonempty intersection  $K_s \cap ((a_{j-1}, a_j) \times R)$  we may combine all the exponential factors

associated to  $K_s$  in this way into one factor  $\exp[-|K_s| E^{(s)}]$ . Upon carrying this out for all  $s$  all the exponential factors in all the Bralic maps are exhausted by Corollary 4.8.

## 7. INVARIANCE PROPERTIES OF EXPECTATIONS

Using the results established in the previous sections, we will now analyze the expectation of a general product of Wilson loops. Theorem 6.8 allows us to write this expectation as a product (in some order) of connection operators  $\{\hat{J}_i\}$  and exponential operators  $\{\exp -A_s E^{(s)}\}$ , where  $\{A_s\}$  are the areas of the connected components of the complement of the loops. This expression superficially depends on our gauge choice; a different choice of directions for the coordinate axes in  $\mathbb{R}^2$  leads (in general) to different commutant algebras  $\{\mathcal{B}_{k_r}\}$ , different connection operators, and different area operators (see the statement of Theorem 6.8). Our task in this section is to prove that the expectation does not depend on the coordinate directions chosen. We do this by rewriting the expectation in a standard representation (see Definition 7.21) which displays the expected invariance properties. The result we get implies, in particular, Euclidean invariance of expectations, and factorization over disjoint components.

Our first task is to describe the type of curves to which our results apply.

**DEFINITION 7.1.** A curve is a continuous piecewise  $C^1$  function  $\sigma: [0, 1] = I \rightarrow \mathbb{R}^2$ . The image of  $\sigma$  will be denoted  $\sigma(I)$ . The curve  $\sigma$  is closed if  $\sigma(0) = \sigma(1)$ , and  $\sigma$  is simple if  $\sigma(t_1) \neq \sigma(t_2)$ , for all  $0 \leq t_1 < t_2 \leq 1$  unless  $t_1 = 0$  and  $t_2 = 1$ .

We will consider only curves with tangent vectors defined and nonzero at all but a finite number of points in  $\mathbb{R}^2$ . The tangent vector to  $\sigma$  will be denoted  $\sigma'(t)$ .

**DEFINITION 7.2.** Given an interval  $A \subset \mathbb{R}$ , denote by  $A \times \mathbb{R}$  the cylinder set  $\{(x, y) \mid x \in A\}$ . The curve  $\sigma$  is a finite union of graphs over the interval  $A$  if

$$\sigma(I) \cap (A \times \mathbb{R}) = \bigcup_{i=1}^n \sigma(I_i),$$

where  $\{I_i\}$ ,  $1 \leq i \leq n$ , is a collection of disjoint intervals in  $[0, 1]$ , and each set  $\sigma(I_i)$  is the graph of a continuous, piecewise  $C^1$  function on  $A$ . The curve  $\sigma$  is a finite union of graphs if there exist  $a_0 < a_1 < \dots < a_q$  such that  $\sigma(I) \subset [a_0, a_q] \times \mathbb{R}$ , and  $\sigma$  is a finite union of graphs over each interval  $(a_{r-1}, a_r)$ , for  $1 \leq r \leq q$ .

As an example of Definition 7.2, we display a closed curve in Fig. 7.1 and indicate the intervals over which  $\sigma$  is a finite union of graphs.

Next we define operations on curves, including the definition of inverse and an extended product for two curves.

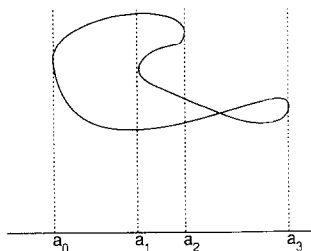


FIG. 7.1. A closed curve which is a finite union of graphs.

DEFINITION 7.3. The inverse of the curve  $\sigma$  is the curve  $\sigma^{-1}$  defined by

$$\sigma^{-1}(t) = \sigma(1 - t), \quad 0 \leq t \leq 1.$$

DEFINITION 7.4. (a) Let  $\sigma_i(t) = (x_i(t), y_i(t))$ ,  $i = 1, 2$ , be two curves satisfying  $x_1(1) = x_2(0)$ . We define the extended product of  $\sigma_1$  and  $\sigma_2$  as the curve

$$\sigma_2 * \sigma_1(t) = \begin{cases} \sigma_1(4t), & 0 \leq t \leq \frac{1}{4} \\ (x_1(1), (2t - \frac{1}{2}) y_2(0) - (2t - \frac{3}{2}) y_1(1)), & \frac{1}{4} \leq t \leq \frac{3}{4} \\ \sigma_2(4t - 3), & \frac{3}{4} \leq t \leq 1. \end{cases}$$

(b) Let  $\sigma(t) = (x(t), y(t))$  satisfy  $x(1) = x(0)$ . We define the closing of  $\sigma$  to be the closed curve

$$* \sigma(t) = \begin{cases} \sigma(2t), & 0 \leq t \leq \frac{1}{2} \\ (x(1), (2t - 1) y(0) - (2t - 2) y(1)), & \frac{1}{2} \leq t \leq 1. \end{cases}$$

*Remark 7.5.* In this last definition, we consider a curve whose endpoints have the same  $x$ -coordinate. By adding on a vertical curve connecting these endpoints, we obtain a closed curve. This construction will be useful later when we define the standard representation of an expectation.

In Section 2, we defined parallel transport along any curve whose image is the graph of a continuous function. Any curve  $\sigma$  which is a finite union of graphs can be written (up to change of parametrization) as the product of a finite number of curves,

$$\sigma = \sigma_n * \sigma_{n-1} * \cdots * \sigma_1,$$

where  $\sigma_i(I)$  is either the graph of a function, or else has constant  $x$ -coordinate, for  $1 \leq i \leq n$ . Parallel transport is trivial in the latter case. We then define parallel transport along  $\sigma$  as

$$U(\sigma) = U(\sigma_n) U(\sigma_{n-1}) \cdots U(\sigma_1).$$

It is important to notice that this definition is independent of the parametrization

of the curve  $\sigma$ . Accordingly, when discussing parallel transport we will not mention parametrization.

We will denote by  $W = \{\sigma_1, \dots, \sigma_n\}$  a collection of curves, and by  $W(I)$  the union of the images of these curves.

**DEFINITION 7.6.** Let  $W = \{\sigma_1, \dots, \sigma_n\}$  be a collection of curves. The point  $p \in W(I)$  is an intersection point of  $W$  if any of the following equations is satisfied:

$$\begin{aligned} p = \sigma_i(t) = \sigma_i(t') & \quad \text{some } 0 < t < t' < 1, \text{ some } 1 \leq i \leq n \\ p = \sigma_i(t) = \sigma_j(t') & \quad \text{some } 0 \leq t \neq t' \leq 1, \text{ some } 1 \leq i < j \leq n. \end{aligned}$$

**DEFINITION 7.7.** The finite collection of closed curves  $W$  is admissible if  $W$  has a finite number of intersection points, and if each curve in  $W$  is a finite union of graphs. The finite collection  $W$  is weakly admissible if  $W(I)^c$  has a finite number of connected components, and if each curve in  $W$  is a finite union of graphs.

*Remark 7.8.* The complement of the image of an admissible collection of closed curves is a finite union of disjoint, connected open sets, one of which is unbounded, and the rest of which are bounded. Thus an admissible collection is weakly admissible (the converse need not be true).

**DEFINITION 7.9.** Let  $W = \{\sigma_1, \dots, \sigma_n\}$  be a weakly admissible collection of closed curves. We define the expectation of  $W$  to be

$$\langle W \rangle = E \left[ \prod_{i=1}^n \text{tr } U(\sigma_i) \right].$$

*Remark 7.10.* Since  $W$  is weakly admissible, there exist  $a_0 < a_1 < \dots < a_q$  such that each curve in  $W$  is a finite union of graphs over each interval  $(a_{r-1}, a_r)$ , for  $1 \leq r \leq q$ , and  $W(I)$  is a subset of  $[a_0, a_q] \times \mathbb{R}$ . Correspondingly, there is a Bralic map  $\varphi_r$  for each interval  $(a_{r-1}, a_r)$ , and a connection operator  $\hat{J}_r$  for each endpoint  $a_r$ . Theorem 5.7 and Remark 5.8 imply that the expectation of  $W$  is given by

$$\langle W \rangle = \hat{J}_0 \varphi_1 \hat{J}_1 \varphi_2 \dots \hat{J}_{q-1} \varphi_q \hat{J}_q.$$

This expression implies invariance of  $\langle W \rangle$  under translations in the  $x$ -direction, since the operators  $\{\varphi_r\}$  and  $\{\hat{J}_r\}$  are invariant under translations in the  $x$ -direction. Furthermore, suppose that  $W = W_1 \cup W_2$ , where  $W_1$  and  $W_2$  are collections of curves whose images  $W_1(I)$  and  $W_2(I)$  are disjoint, and suppose that there is a vertical line which is disjoint from  $W(I)$ , such that  $W_1(I)$  and  $W_2(I)$  lie on different sides of the line. Let  $(a, 0)$  be the point where this line intersects the  $x$ -axis, and suppose that  $a \in (a_{r-1}, a_r)$ . Then the connection operators at the left



and right hand ends of the interval  $(a_{r-1}, a_r)$ , namely  $\hat{J}_{r-1}$  and  $\hat{J}_r$ , have the algebra  $\mathcal{B}_0 = \mathbb{C}$  as domain and range, respectively. Remark 5.8 implies that

$$\begin{aligned}\langle W_1 \rangle &= \hat{J}_0 \varphi_1 \hat{J}_1 \cdots \varphi_{r-1} \hat{J}_{r-1} \\ \langle W_2 \rangle &= \hat{J}_r \varphi_{r+1} \cdots \varphi_q \hat{J}_q.\end{aligned}$$

Since  $\varphi_r$  is equal to one, this implies

$$\langle W \rangle = \langle W_1 \rangle \langle W_2 \rangle.$$

We will now analyze the expectation of an admissible collection  $W$ , and show that it equals the expectation of another collection, called a standard representation. The following definitions will provide an inductive procedure for finding this standard representation.

**DEFINITION 7.11.** Let  $W$  be an admissible collection of closed curves, and let  $\{R_0, R_1, \dots, R_k\}$  be the connected components of  $W(I)^c$ , where  $R_0$  is the unbounded component. We associate to this a graph  $V(W)$  on  $k+1$  vertices  $\{v_0, v_1, \dots, v_k\}$ , with edges connecting each pair of vertices  $v_i$  and  $v_j$  for which  $\partial R_i \cap \partial R_j$  contains a nonempty open subset of  $W(I)$ .

*Remark 7.12.* For an admissible collection  $W$ , the associated graph  $V(W)$  is connected. Fig. 7.2 shows several examples of collections with their associated graphs.

**DEFINITION 7.13.** Let  $V$  be a connected graph. A tree on  $V$  is a minimal connected subset of  $V$  which contains every vertex. The distance between two vertices  $v$  and  $v'$  in  $T$  is the number of edges in the smallest connected subset of  $T$  containing  $v$  and  $v'$ .

With the choice of a base vertex, a tree becomes a partially ordered set, in the

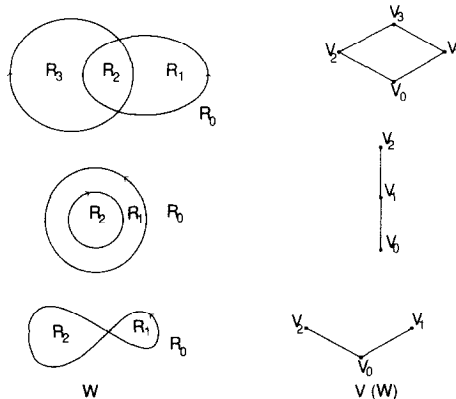


FIG. 7.2. Collections of curves with their associated graphs.

following way. Suppose  $T$  is a tree on  $V(W)$ , and the base vertex is  $v_0$ . Then there is a unique sequence in  $T$  from  $v_0$  to  $v_m$ , for each  $1 \leq m \leq k$ . We define a partial order on the vertices in  $T$  by saying  $v_i < v_m$  if  $v_i$  belongs to the sequence connecting  $v_0$  to  $v_m$ . So, for example,  $v_0 < v_m$  for all  $1 \leq m \leq k$ . The set of vertices  $\{v_i \mid v_i < v_m\}$  is an ordered set, and we define  $v_{\bar{m}}$  as the (unique) maximum element of this set, for  $1 \leq m \leq k$ .

Now let  $W = \{\sigma_1, \dots, \sigma_n\}$  be an admissible collection of closed curves. By translation invariance in the  $x$ -direction, we can assume that  $W(I)$  lies to the left of the  $y$ -axis. Let  $T$  be a tree on  $V(W)$ . For each pair of vertices  $v_i, v_j$  connected by an edge in  $T$ , choose a point in  $\partial R_i \cap \partial R_j$  which is not an intersection point of  $W$ , and at which the tangent vector exists and is nonzero. If  $v_i > v_j$ , call this point  $p_i$ ; otherwise call it  $p_j$ . Define  $p_0$  to be the origin in the plane. We will call  $p_i$  the gate for the region  $R_i$ , and we will denote by  $T_i$  the tangent vector to the parametrized curve at  $p_i$ . Note that each region  $R_m$  has at least one gate on its boundary, namely  $p_m$ , for  $1 \leq m \leq k$ . If  $p_j$  is another gate on the boundary, then  $j = m$ .

**DEFINITION 7.14.** Let  $T$  be a tree on  $V(W)$  and let  $\{p_m\}$  be gates for the regions  $\{R_m\}$ . For each  $m = 1, \dots, k$ , let  $\Gamma_m$  be a simple  $C^1$  curve with  $\Gamma_m(0) = p_m$  and  $\Gamma_m(1) = p_{\bar{m}}$ , such that  $\Gamma_m(t) \in R_{\bar{m}}$  for  $0 < t < 1$ . If  $\bar{m} = 0$ , we require that  $\Gamma_m(t) \in R_0 \cap \{(x, y) \mid x < 0\}$  for  $0 < t < 1$ . We also require that at each gate  $p_m$ , the vectors  $T_m$ ,  $\Gamma'_m(0)$  and  $\{\Gamma'_j(1) \mid j = m\}$  should be non-zero, and no two of these vectors should be parallel. Let  $\Gamma$  be the collection of these curves. Then we say that  $\Gamma$  is a tree on  $W$  if  $\Gamma$  has no intersection points other than the gates  $\{p_0, p_1, \dots, p_k\}$ , and if each curve  $\Gamma_m$  is a finite union of graphs.

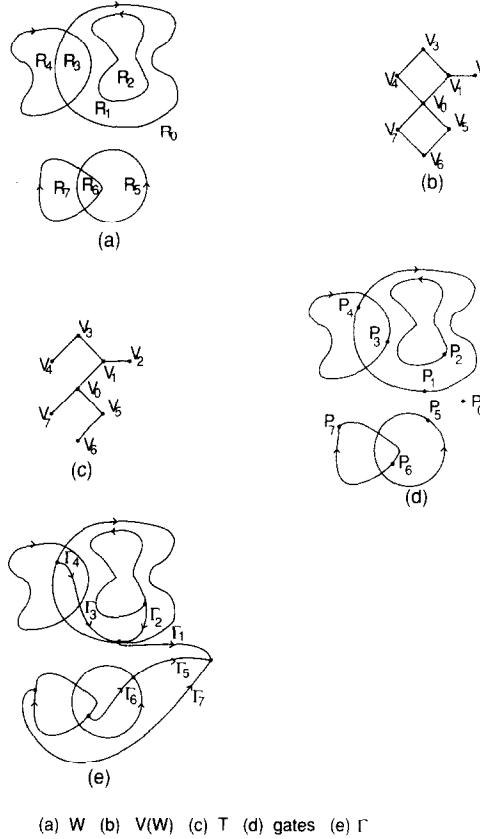
*Remark 7.15.* The existence of these curves  $\{\Gamma_m\}$  follows from our assumptions on  $W$ , as explained in Lemma A1 in the Appendix. Fig. 7.3 shows an example of a tree  $\Gamma$  on a collection  $W$ . We will use  $\Gamma$  to extract from  $W$  some information about how the regions  $\{R_0, R_1, \dots, R_k\}$  are arranged relative to the curves in  $W$ . It will turn out that this information, along with the areas of the components  $\{R_m\}$ , will completely determine  $\langle W \rangle$ . First we recall the following definition.

**DEFINITION 7.16.** Let  $v = (v_1, v_2)$  and  $w = (w_1, w_2)$  be two vectors given by their cartesian coordinates. We define

$$\text{sgn}(v, w) = \text{sgn}(v_1 w_2 - v_2 w_1).$$

**DEFINITION 7.17.** For each  $m, j = 1, \dots, k$ , we define

$$\begin{aligned} \delta_m &= \text{sgn}(\Gamma'_m(0), T_m) \\ \delta(m, j) &= \begin{cases} \text{sgn}(\Gamma'_m(1), \Gamma'_j(1)) & \text{if } \bar{m} = j, m \neq j \neq 0 \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$


 FIG. 7.3. Constructing a tree  $\Gamma$  as a collection of curves  $W$ .

**DEFINITION 7.18.** We shall say that the  $2k$  distinct numbers  $\{a_1, \dots, a_k; b_1, \dots, b_k\}$  are ordered according to  $\Gamma$  if the following conditions are satisfied:

- (i)  $a_m < b_m$ ,  $1 \leq m \leq k$
- (ii)  $a_m < a_j < b_j < b_m$  if  $v_m < v_j$
- (iii)  $a_j < b_j < a_m < b_m$  if  $\delta(m, j) = 1$ .

**Remark 7.19.** These conditions are sufficient to determine the ordering of the  $2k$  numbers  $\{a_1, \dots, a_k; b_1, \dots, b_k\}$ . The relevance of Definition 7.18 will become clear when we define the standard representation for  $\langle W \rangle$ . Our standard representation uses  $k$  non-intersecting curves in the right half-plane with endpoints lying on the  $y$ -axis. The curves will enclose regions corresponding to the bounded components of  $W(I)^c$ , and their endpoints on the  $y$ -axis will be ordered according to  $\Gamma$ . As an example, we present below the ordering obtained from the example in Fig. 7.3.

EXAMPLE 7.20. From Fig. 7.3, we get the following:

$$\delta_2 = \delta_4 = \delta_5 = 1; \delta_1 = \delta_3 = \delta_6 = \delta_7 = -1$$

$$\delta(2, 3) = \delta(1, 5) = \delta(1, 7) = \delta(5, 7) = 1$$

$$\delta(3, 2) = \delta(5, 1) = \delta(7, 1) = \delta(7, 5) = -1$$

$$\delta(m, j) = 0 \quad \text{otherwise.}$$

This gives the following ordering on  $\{a_1, \dots, a_7; b_1, \dots, b_7\}$ :

$$a_7 < b_7 < a_5 < a_6 < b_6 < b_5 < a_1 < a_3 < a_4 < b_4 < b_3 < a_2 < b_2 < b_1.$$

The final piece of information we need concerns the relative positions of the gates on the curves. Each gate belongs to a curve  $\sigma_i$  in  $W$ , and the curve  $\sigma_i$  gives an ordering for the gates belonging to it. If  $0 \leq t_1 < t_2 \leq 1$ , we will write  $\sigma(t_1) < \sigma(t_2)$  to indicate that  $\sigma(t_1)$  precedes  $\sigma(t_2)$ . Let  $g(i)$  be the number of gates in  $\sigma_i(I)$ . We list the gates in  $\sigma_i(I)$  in order along the curve, as  $p_{i_1} < p_{i_2} < \dots < p_{i_{g(i)}}$ . Correspondingly, let  $\theta_{i,j}$  be the part of the curve  $\sigma_i$  connecting two adjacent gates  $p_{i_j}$  and  $p_{i_{j+1}}$  ( $\theta_{i,g(i)}$  connects  $p_{i_{g(i)}}$  and  $p_{i_1}$ ). Then the curve  $\sigma_i$  can be written as the product

$$\sigma_i = \theta_{i,g(i)} * \theta_{i,g(i)-1} * \dots * \theta_{i,1}, \quad 1 \leq i \leq n \quad (7.1)$$

(here we have reparametrized  $\sigma_i$  so that it starts at  $p_{i_1}$ ). We can now define the standard representation, using the information described above. We will denote by  $\text{int}(\sigma)$  the union of the bounded connected components of  $\sigma(I)^c$ , for any closed curve  $\sigma$ .

DEFINITION 7.21. Let  $\{a_1, \dots, a_k; b_1, \dots, b_k\}$  be ordered according to  $\Gamma$ , and let  $\{w_m = (0, a_m); z_m = (0, b_m)\}$ ,  $1 \leq m \leq k$ , be the corresponding  $2k$  points on the  $y$ -axis. For each  $m = 1, \dots, k$ , let  $\gamma_m$  be a simple  $C^1$  curve with  $\gamma_m(0) = w_m$  and  $\gamma_m(1) = z_m$ , and with  $\gamma_m(t) \in \{(x, y) \mid x > 0\}$  for all  $0 < t < 1$ . Define  $n$  closed curves as follows (see Definitions 7.3, 7.4):

$$\bar{\sigma}_i = * \gamma_{i_{g(i)}}^{\delta_{i_{g(i)}}} * \dots * \gamma_{i_2}^{\delta_{i_2}} * \gamma_{i_1}^{\delta_{i_1}}, \quad 1 \leq i \leq n \quad (7.2)$$

(where  $\gamma^\delta = \gamma^{\pm 1}$  is either  $\gamma$  or its inverse). We will call the collection of curves  $\bar{W} = \{\bar{\sigma}_1, \dots, \bar{\sigma}_n\}$  a standard representation for  $W$  if the following conditions hold:

- (i) each curve  $\bar{\sigma}_i$  is a finite union of graphs
- (ii) the only intersection points of  $\bar{W}$  lie on the  $y$ -axis and
- (iii)

$$\left| \text{int}(*\gamma_m) \setminus \bigcup_{v_j > v_m} \text{int}(*\gamma_j) \right| = |R_m|, \quad 1 \leq m \leq k. \quad (7.3)$$

*Remark 7.22.* Our standard representation should be seen as resulting from a particular deformation of the curves in  $W$  which preserves the areas of the components of the complement. Each curve  $\sigma_i$  is deformed into  $\bar{\sigma}_i$ , and all the intersection points of the curves end up on the  $y$ -axis. The condition (7.3) ensures that the areas of components of  $W(I)^c$  are preserved. Figure 7.4 illustrates the construction of a standard representation for the collection of curves in Fig. 7.3.

We now have our main invariance result, which states that  $\langle W \rangle$  can be computed from the information recorded in the tree  $\Gamma$ , the ordering of gates around curves, and the areas of the components  $\{R_1, \dots, R_k\}$ .

**THEOREM 7.23.** *Let  $W$  be an admissible collection of closed curves, and let  $\bar{W}$  be a standard representation of  $W$ . Then*

$$\langle W \rangle = \langle \bar{W} \rangle.$$

In order to prove Theorem 7.23, we will construct a family of collections of curves  $\{\bar{W}(\alpha)\}$  which interpolate between  $W$  and  $\bar{W}$ , for  $0 \leq \alpha \leq 1$ . This will involve the following definition of a squashed curve.

**DEFINITION 7.24.** Let  $\sigma(t) = (x(t), y(t))$  be a curve, and let  $\alpha$  satisfy  $0 \leq \alpha \leq 1$ . Then we define the squashed curve  $\sigma(\alpha)$  by

$$\sigma(\alpha; t) = (x(t), \alpha y(t)).$$

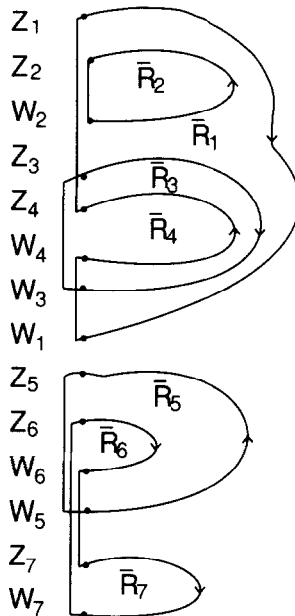


FIG. 7.4. Standard representation for the curves in Fig. 7.3. The region corresponding to  $R_m$  has been denoted  $\bar{R}_m$ , for  $m = 1, \dots, 7$ .

Note that if  $\sigma$  is the graph of a function over the interval  $(a, b)$ , then so also is  $\sigma(\alpha)$ , for  $0 \leq \alpha \leq 1$ .

Now let  $\Gamma$  be a tree on  $W$ . By taking products of curves in  $\Gamma$ , we can obtain a simple curve  $\tilde{\Gamma}_m$  connecting  $p_m$  to  $p_0$ , for each  $m = 1, \dots, k$ . (So  $\tilde{\Gamma}_m(I)$  is the minimal connected subset of  $\Gamma(I)$  which contains  $p_m$  and  $p_0$ .) Then we define an extension of the curve  $\sigma_i$  in the following way (recall that the definition of  $\bar{W}$  involves the curves  $\{\gamma_1, \dots, \gamma_k\}$ ):

$$\begin{aligned} \tilde{\sigma}_i = & \theta_{i, g(i)} * \tilde{\Gamma}_{i, g(i)}^{-1} * \gamma_{i, g(i)}^{\delta_{i, g(i)}} * \tilde{\Gamma}_{i, g(i)} * \dots \\ & * \theta_{i, 2} * \tilde{\Gamma}_{i, 2}^{-1} * \gamma_{i, 2}^{\delta_{i, 2}} * \tilde{\Gamma}_{i, 2} * \theta_{i, 1} * \tilde{\Gamma}_{i, 1}^{-1} * \gamma_{i, 1}^{\delta_{i, 1}} * \tilde{\Gamma}_{i, 1}, \quad 1 \leq i \leq n. \end{aligned} \quad (7.4)$$

The reader should compare Eqs. (7.4) and (7.1). As an example. Fig. 7.5 illustrates the case of two curves  $\sigma_1$  and  $\sigma_2$ , along with their extensions  $\tilde{\sigma}_1$  and  $\tilde{\sigma}_2$ . The curves have been separated so that they can be distinguished easily. In fact, we will prove our results by approximating the curves  $\{\tilde{\sigma}_i\}$  in just this way, so that the collections will be admissible. By taking a limit where the width of the “tubes” goes to zero, we will recover the result for the curves defined by Eq. (7.4). Furthermore, the reader can check that the vertical ordering of the loops  $\gamma_1, \gamma_2$ , and  $\gamma_3$  in Fig. 7.5 is determined by the tree  $\Gamma$  in accordance with Definition 7.18. This should serve as a motivation for our definitions, and should help the interested reader to construct a standard representation for any other collection of curves.

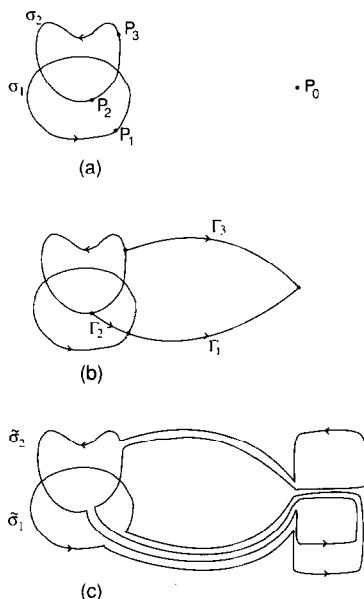


FIG. 7.5. Construction of interpolating curves.

Now we use Definition 7.24 to squash this curve in the following way. For  $0 \leq \alpha \leq 1$ , define the curve

$$\begin{aligned} \tilde{\sigma}_{i,\alpha} = & \theta_{i,g(i)}(1-\alpha) * \tilde{F}_{i_{g(i)}}^{-1}(1-\alpha) * \gamma_{i_{g(i)}}^{\delta_{i_{g(i)}}}(\alpha) * \tilde{F}_{i_{g(i)}}(1-\alpha) \\ & * \cdots * \theta_{i,1}(1-\alpha) * \tilde{F}_{i_1}^{-1}(1-\alpha) * \gamma_{i_1}^{\delta_{i_1}}(\alpha) * \tilde{F}_{i_1}(1-\alpha). \end{aligned} \quad (7.5)$$

Note that every curve to the left of the  $y$ -axis has been squashed by a factor  $1-\alpha$ , while every curve to the right of the  $y$ -axis has been squashed by a factor  $\alpha$ . This is designed to keep fixed the areas of the connected components of the complement of the curves. For example in Fig. 7.5, if we neglect the areas contained in the "tubes", each connected component (there are only three) has equal areas on the right and left of the  $y$ -axis. When the curves are squashed according to Eq. (7.5), the total area of each connected component does not change as a function of  $\alpha$ . In the limiting case  $\alpha=0$ , the reader can check that parallel transport around the curves  $\tilde{\sigma}_{1,0}$  and  $\tilde{\sigma}_{2,0}$  is the same as for the original curves  $\sigma_1$  and  $\sigma_2$ . Similarly, when  $\alpha=1$ , we get the parallel transport for the curves  $\tilde{\sigma}_1$  and  $\tilde{\sigma}_2$ . We will prove that the expectation does not change as the curves are squashed in this way, and this will imply Theorem 7.23.

Now let  $\tilde{W}(\alpha)$  be the collection of closed curves  $\{\tilde{\alpha}_{1,\alpha}, \dots, \tilde{\sigma}_{n,\alpha}\}$ , for each  $0 \leq \alpha \leq 1$ . Note that  $\tilde{W}(\alpha)$  is a weakly admissible collection of closed curves (it is not admissible since the intersection points form an open set).

**PROPOSITION 7.25.**  $\langle \tilde{W}(\alpha) \rangle$  is independent of  $\alpha$ , for  $0 \leq \alpha \leq 1$ .

Before proving Proposition 7.25, we use it to deduce Theorem 7.23, and we also deduce our main results about invariance properties of expectations.

*Proof of Theorem 7.23.* By Proposition 7.25 we have

$$\langle \tilde{W}(0) \rangle = \langle \tilde{W}(1) \rangle.$$

When  $\alpha=0$ , the squashed curve  $\gamma_i(\alpha)$  lies entirely on the  $x$ -axis, where parallel transport is trivial. Therefore (7.1) and (7.5) imply that parallel transport along  $\tilde{\sigma}_{i,0}$  is given by

$$\begin{aligned} U(\tilde{\sigma}_{i,0}) &= U(\theta_{i,g(i)}) \cdots U(\theta_{i,1}) \\ &= U(\sigma_i). \end{aligned}$$

Hence we deduce

$$\langle \tilde{W}(0) \rangle = \langle W \rangle.$$

Furthermore, when  $\alpha=1$ , (7.2) and (7.5) imply that

$$\begin{aligned} U(\tilde{\sigma}_{i,1}) &= U(\gamma_{i_{g(i)}}^{\delta_{i_{g(i)}}}) \cdots U(\gamma_{i_1}^{\delta_{i_1}}) \\ &= U(\tilde{\sigma}_i) \end{aligned}$$

and hence

$$\langle \tilde{W}(1) \rangle = \langle \tilde{W} \rangle.$$

If  $W = \{\sigma_1, \dots, \sigma_n\}$  is a collection of curves and  $\varphi: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  is a  $C^1$  mapping, we will denote by  $W_\varphi$  the collection of curves  $\{\varphi \circ \sigma_1, \dots, \varphi \circ \sigma_n\}$ .

**THEOREM 7.26.** *Let  $W = \{\sigma_1, \dots, \sigma_n\}$  be an admissible collection of closed curves, and let  $\varphi: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  be an orientation-preserving diffeomorphism. Let  $\{R_1, \dots, R_k\}$  be the bounded connected components of  $W(I)^c$ , and let  $\{\varphi(R_1), \dots, \varphi(R_k)\}$  be the corresponding components of  $W_\varphi(I)^c$ . Suppose that  $W_\varphi$  is an admissible collection and that*

$$|R_m| = |\varphi(R_m)|, \quad 1 \leq m \leq k.$$

Then

$$\langle W \rangle = \langle W_\varphi \rangle.$$

*Proof.* Let  $\Gamma$  be a tree on  $W$  (see Definition 7.14), and let  $\varphi \circ \Gamma$  be the image of  $\Gamma$  under  $\varphi$ . By translation in the  $x$ -direction, if necessary, we can assume that  $W_\varphi(I)$  and  $\varphi \circ \Gamma(I)$  lie to the left of the  $y$ -axis (see Remark 7.10). By adding on a  $C^1$  curve connecting the translated endpoint of  $\varphi \circ \Gamma$  to the origin, and applying Lemma A3 from the Appendix, we obtain a tree  $\Gamma_\varphi$  on  $W_\varphi$ . Since  $\varphi$  is orientation preserving, the values of the coefficients  $\{\delta_m\}$  and  $\{\delta(m, j)\}$  are preserved under  $\varphi$ , for each  $m, j = 1, \dots, k$ . Therefore  $\Gamma$  and  $\Gamma_\varphi$  define the same ordering for  $2k$  points  $\{a_1, \dots, a_k; b_1, \dots, b_k\}$ . Furthermore the ordering of gates on each curve  $\sigma_i$  is preserved under  $\varphi$ . Therefore since the areas of the corresponding connected components in  $W(I)^c$  and  $W_\varphi(I)^c$  are equal, it follows that  $W$  and  $W_\varphi$  share the same standard representations. Hence Theorem 7.23 implies that  $\langle W \rangle = \langle W_\varphi \rangle$ .

**COROLLARY 7.27** (Euclidean invariance). *Let  $\varphi: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  be a rotation or translation. Let  $W$  be an admissible collection, and suppose  $W_\varphi$  is also admissible. Then*

$$\langle W \rangle = \langle W_\varphi \rangle.$$

**COROLLARY 7.28** (Factorization of components). *Let  $W$  be an admissible collection of closed curves. Let  $\varphi: \mathbb{R} \rightarrow \mathbb{R}^2$  be a continuous, simple curve satisfying  $\varphi(t) = (t, 0)$  for  $|t| \geq T$ , for some  $T > 0$ , such that the image of  $\varphi$  is disjoint from  $W(I)$ . Let  $W_1$  and  $W_2$  be the subcollections of curves in  $W$  whose images  $W_1(I)$  and  $W_2(I)$  belong to different components of the complement of the image of  $\varphi$ . Then*

$$\langle W \rangle = \langle W_1 \rangle \langle W_2 \rangle.$$

*Proof.* The proof follows by choosing an appropriate tree  $\Gamma$  on  $W$  and applying Remark 7.10. By translation invariance, we can translate  $W$  and  $\varphi$  so that  $W(I)$  lies



to the left of the  $y$ -axis, and so that the translation of the curve  $\varphi$  passes through the origin in the plane. The translated curve  $\varphi$  divides the plane into two components  $N_1$  and  $N_2$ , where  $N_1$  lies above the  $x$ -axis asymptotically, and  $N_2$  lies below. Without loss of generality, we can assume  $W_1(I) \subset N_1$ , and so we can choose the curves in  $\Gamma$  attached to  $W_1$  so that their images all lie in  $N_1$  also (excluding endpoints at the origin). Similarly the curves in  $\Gamma$  attached to  $W_2$  can be chosen to lie in  $N_2$ , except for the endpoints at the origin. With this choice of  $\Gamma$ , we can find a standard representation  $\bar{W}$  for which the curves associated with  $W_1$  all lie above the  $x$ -axis, and the curves associated with  $W_2$  all lie below. By rotation invariance, we can rotate the curves in  $\bar{W}$  through  $90^\circ$ , in which case the  $y$ -axis separates the curves into disjoint sets. Since the expectation factorizes in this case (see Remark 7.10), it also factorizes for  $\bar{W}$  and hence for  $W$ .

*Proof of Proposition 7.25.* We will construct a sequence of admissible collections of curves  $\tilde{W}(\alpha; \varepsilon)$  depending on a parameter  $\varepsilon$ , which converge pointwise to  $\tilde{W}(\alpha)$  as  $\varepsilon \rightarrow 0$ . For each value of  $\varepsilon$ , we will prove that  $\langle \tilde{W}(\alpha; \varepsilon) \rangle$  is independent of  $\alpha$ , and therefore  $\langle \tilde{W}(\alpha) \rangle$  also will be independent of  $\alpha$  (see Remark 4.10). Figure 7.5 illustrates how these approximations are constructed, namely by replacing the curves in the tree  $\Gamma$  by non-intersecting “tubes”. Each region  $R_m$  in  $W(I)^c$  is connected by a tube to a region of equal area on the right of the  $y$ -axis. This new collection of curves is admissible, and after squashing by a factor  $1 - \alpha$  on the left of the  $y$ -axis and by  $\alpha$  on the right, we obtain the collection  $\tilde{W}(\alpha; \varepsilon)$ . Except for some overlap of lines on the  $y$ -axis, the collection  $\tilde{W}(\alpha; \varepsilon)$  is also admissible. Therefore by the result of Section 5, we can write  $\langle \tilde{W}(\alpha; \varepsilon) \rangle$  as a product of Bralic maps and connection operators. First we will show that the dependence on  $\alpha$  occurs only in the Bralic maps  $\{\varphi_r\}$ . Then we will show that these Bralic maps may be combined in such a way that the dependence on  $\alpha$  disappears (this uses the results of Section 6). It is important to notice that the vertical ordering of points on the  $y$ -axis produced by  $\Gamma$  is exactly that which allows us to construct disjoint tubes connecting  $W$  with the standard representation.

Now let  $\varepsilon > 0$ , and let  $B_\varepsilon(p)$  be the disk of radius  $\varepsilon$  centered at  $p$ . We will assume  $\varepsilon$  is sufficiently small so that the disks about the gates  $\{B_\varepsilon(p_m)\}$ ,  $0 \leq m \leq k$ , are disjoint. For each  $m = 1, \dots, k$ , choose two points  $q_m, r_m$  in  $B_\varepsilon(p_m)$  which lie on the curve containing  $p_m$ , and satisfy  $q_m < p_m < r_m$  (see remarks after Example 7.20). We will denote by  $\psi_m$  the part of the curve in  $W$  which connects  $q_m$  to  $r_m$ , and passes through  $p_m$ . Furthermore, for any two adjacent gates  $p_{i_j}$  and  $p_{i_{j+1}}$  on a curve  $\sigma_i$  in  $W$ , let  $\chi_{i,j}$  be the part of  $\sigma_i$  connecting  $r_{i_j}$  to  $q_{i_{j+1}}$ . Then we have the following representation of  $\sigma_i$  (compare with (7.1)):

$$\sigma_i = \chi_{i, g(i)} * \psi_{i_{g(i)}} * \chi_{i, g(i)-1} * \dots * \chi_{i, 1} * \psi_{i_1}, \quad 1 \leq i \leq n. \quad (7.6)$$

We now use the following result, which makes concrete the “tubes” illustrated in Fig. 7.5. The result says that each curve  $\tilde{\Gamma}_m$  (which connects  $p_m$  to  $p_0$ , the origin) can be replaced by a tube bounded by two curves  $\rho_m$  and  $\tau_m$ , in such a way that all the new curves  $\{\rho_m, \tau_m\}$  are disjoint.

LEMMA 7.29. *There are  $2k$  distinct points  $\{w'_m, z'_m\}$ ,  $m=1, \dots, k$ , on the  $y$ -axis within distance  $\varepsilon$  of the origin, whose  $y$ -coordinates are ordered according to  $\Gamma$ , and  $2k$  curves  $\{\rho_m, \tau_m\}$ ,  $m=1, \dots, k$ , satisfying the following conditions:*

- (i)  $\rho_m(1) = w'_m$ ,  $\tau_m(1) = z'_m$
- (ii)  $\rho_m(0) = q_m$ ,  $\tau_m(0) = r_m$  if  $\delta_m = 1$ ;  $\rho_m(0) = r_m$ ,  $\tau_m(0) = q_m$  if  $\delta_m = -1$
- (iii) *excluding endpoints, the curves  $\{\rho_m, \tau_m; \chi_{i,j}\}$  are disjoint, for all  $1 \leq m, m' \leq k$ ,  $1 \leq i \leq n$ , and  $1 \leq j \leq g(i)$ , each curve is a finite union of graphs, and each curve lies to the left of the  $y$ -axis*
- (iv) *the images  $\rho_m(I)$  and  $\tau_m(I)$  belong to the  $\varepsilon$ -neighborhood of the set  $\tilde{\Gamma}_m(I)$ , for each  $1 \leq m \leq k$ .*

This lemma is proved in the Appendix; we will now use it to establish Proposition 7.25. Let  $\bar{W}$  be a standard representation for  $W$ , so that  $\bar{W}$  is a collection of  $n$  closed curves  $\{\bar{\sigma}_1, \dots, \bar{\sigma}_n\}$  built out of the  $k$  curves  $\{\gamma_1, \dots, \gamma_k\}$ . We will consider a different set of curves  $\{\beta_1, \dots, \beta_k\}$  which closely approximate  $\{\gamma_1, \dots, \gamma_k\}$ . They are constructed according to Definition 7.21, but with the area condition (iii) slightly modified. We shall specify the corresponding condition shortly. Then for each  $m=1, \dots, k$  we define the curve

$$C_m = \tau_m^{-1} * \beta_m * \rho_m. \quad (7.7)$$

Using these curves, we construct  $n$  closed curves  $\{\mu_1, \dots, \mu_n\}$  as follows:

$$\begin{aligned} \mu_i &= \chi_{i, g(i)} * C_{i, g(i)}^{\delta_{i, g(i)}} * \chi_{i, g(i)-1} * \dots * \chi_{i, 2} \\ &\quad * C_{i, 2}^{\delta_{i, 2}} * \chi_{i, 1} * C_{i, 1}^{\delta_{i, 1}}, \quad 1 \leq i \leq n. \end{aligned} \quad (7.8)$$

Again  $C^\delta = C^{\pm 1}$  is either the curve  $C$  or its inverse. We will denote by  $\tilde{W}(\varepsilon)$  the weakly admissible collection of curves  $\{\mu_1, \dots, \mu_n\}$ . Next we define squashed curves as before. First we have

$$C_{m, \alpha} = \tau_m^{-1}(1 - \alpha) * \beta_m(\alpha) * \rho_m(1 - \alpha), \quad 1 \leq m \leq k \quad (7.9)$$

for  $0 \leq \alpha \leq 1$ . This then gives us

$$\begin{aligned} \mu_{i, \alpha} &= \chi_{i, g(i)}(1 - \alpha) * C_{i, g(i), \alpha}^{\delta_{i, g(i)}} * \chi_{i, g(i)-1}(1 - \alpha) * \dots \\ &\quad * \chi_{i, 1}(1 - \alpha) * C_{i, 1, \alpha}^{\delta_{i, 1}}, \quad 1 \leq i \leq n. \end{aligned} \quad (7.10)$$

We will denote by  $\tilde{W}(\alpha; \varepsilon)$  the weakly admissible collection of curves  $\{\mu_{1, \alpha}, \dots, \mu_{n, \alpha}\}$ . Consider the part of  $\tilde{W}(\alpha; \varepsilon)(I)^c$  lying to the left of the  $y$ -axis. Each bounded component of this set contains on its boundary an interval on the  $y$ -axis, lying inside the disk of radius  $\varepsilon$  about the origin. The endpoints of these intervals belong to the set of points  $\{w'_m, z'_m\}$ ,  $m=1, \dots, k$ . We will collect these components into sets  $\{K_m\}$ ,  $m=1, \dots, k$ , according to the intervals on their boundaries. For brevity, we will denote by  $[a, b]$  the interval on the  $y$ -axis with endpoints  $a$  and  $b$ ;

where  $a$  lies below  $b$ . For each  $m = 1, \dots, k$ , let  $K_m$  be the union of the connected components whose boundaries contain one of the intervals  $[w'_m, z'_m]$ ,  $[w'_m, w'_j]$ ,  $[z'_j, z'_m]$  or  $[z'_j, w'_i]$  with  $j = i = m$ . These intervals have been chosen so that in the limit  $\varepsilon \rightarrow 0$ , the set  $K_m$  will converge to  $R_m$ , along with some part of the tree  $\Gamma$ . For each  $m$ , the set  $K_m$  contains (say)  $d_m$  connected components  $\{K_{m,1}, \dots, K_{m,d_m}\}$ . We can now define the area constraint for the curve  $\beta_m$ :

$$\left| \text{int}(*\beta_m) / \bigcup_{v_j > v_m} \text{int}(*\beta_j) \right| = \sum_{i=1}^{d_m} |K_{m,i}|, \quad 1 \leq m \leq k. \quad (7.11)$$

In the limit  $\varepsilon \rightarrow 0$ , the area of the set  $K_m$  converges to  $|R_m|$ . It is straightforward to show that we can find curves  $\{\beta_1, \dots, \beta_k\}$  which converge to the original curves  $\{\gamma_1, \dots, \gamma_k\}$  as  $\varepsilon \rightarrow 0$ . In this case the collection of curves  $\tilde{W}(\alpha; \varepsilon)$  converges to  $\tilde{W}(\alpha)$ .

The boundary of the connected set  $\text{int}(*\beta_m) / \bigcup_{v_j > v_m} \text{int}(*\beta_j)$  contains intervals on the  $y$ -axis whose endpoints belong to the set  $\{w_1, \dots, w_k; z_1, \dots, z_k\}$ . These intervals are precisely those described above for  $K_m$ , but with each  $w'_\alpha, z'_\alpha$  replaced by  $w_\alpha, z_\alpha$ , for  $\alpha = i, j, m$ .

We now consider the expectation  $\langle \tilde{W}(\varepsilon) \rangle$ . There is a set of numbers  $c_0 < c_1 < \dots < c_p < 0 < c_{p+1} < \dots < c_{p+q}$  such that  $\tilde{W}(\varepsilon)$  is a finite union of graphs over each interval  $(c_{r-1}, c_r)$ , for  $1 \leq r \leq p+q$ . Therefore we have

$$\langle \tilde{W}(\varepsilon) \rangle = \hat{J}_0 \varphi_1 \hat{J}_1 \varphi_2 \dots \hat{J}_{p-1} \varphi_p \varphi_{p+1} \hat{J}_{p+1} \dots \varphi_{p+q} \hat{J}_{p+q}$$

where  $\{\hat{J}_r, \varphi_r\}$  are the corresponding connection operators and Bralic maps. Note that the connection operator at  $x=0$  is the identity, since the vertical ordering of the curves arriving from both sides is given by the tree  $\Gamma$ . In any interval  $(c_{r-1}, c_r)$ , the Bralic map is a product of exponentials

$$\varphi_r = \exp[-A_{r,1} E_{r,1}] \dots \exp[-A_{r,s} E_{r,s}],$$

where  $\{A_{r,i}\}$  are the areas of connected components, and  $\{E_{r,i}\}$  are the area operators. If each curve in this interval is squashed by a factor  $\alpha$ , the corresponding Bralic map is

$$\varphi_r(\alpha) = \exp[-\alpha A_{r,1} E_{r,1}] \dots \exp[-\alpha A_{r,s} E_{r,s}].$$

Furthermore, the positions of the points  $\{c_0, c_1, \dots, c_{p+q}\}$  and the connection operators  $\{\hat{J}_r\}$  do not change when curves are squashed. Therefore we also have, for  $0 \leq \alpha \leq 1$ ,

$$\begin{aligned} \langle \tilde{W}(\alpha; \varepsilon) \rangle &= \hat{J}_0 \varphi_1(1-\alpha) \hat{J}_1 \varphi_2(1-\alpha) \dots \hat{J}_{p-1} \varphi_p(1-\alpha) \varphi_{p+1}(\alpha) \hat{J}_{p+1} \\ &\quad \dots \varphi_{p+q}(\alpha) \hat{J}_{p+q}. \end{aligned}$$

The limiting cases  $\alpha = 0, 1$  follow from Corollary 4.9 and Remark 4.10 on convergence of expectations. Note that the dependence on  $\alpha$  occurs only in the Bralic maps, and not the connection operators.

Now consider one of the sets  $K_m$  lying to the left of the  $y$ -axis (these were introduced after Eq. (7.10)). It is composed of  $d_m$  connected sets  $\{K_{m,1}, \dots, K_{m,d_m}\}$ , and by construction each component  $K_{m,i}$  intersects the vertical strip  $(c_p, 0) \times \mathbb{R}$ . For each  $K_{m,i}$ , we choose one connected component of  $K_{m,i} \cap ((c_p, 0) \times \mathbb{R})$ , and we let  $E_{m,i}$  be the area operator associated to this component. We define a new Bralic map for the interval  $(c_p, 0)$ , namely,

$$\tilde{\varphi}_p(1 - \alpha) = \exp \left[ -(1 - \alpha) \sum_{m=1}^k \sum_{i=1}^{d_m} |K_{m,i}| E_{m,i} \right].$$

Since the complement of all curves lying to the left of the  $y$ -axis is the union of the sets  $\{K_m\}$ , it follows from Corollary 4.8 and Theorem 6.8 that

$$\langle \tilde{W}(\alpha; \varepsilon) \rangle = \hat{J}_0 \hat{J}_1 \cdots \hat{J}_{p-1} \tilde{\varphi}_p(1 - \alpha) \varphi_{p+1}(\alpha) \hat{J}_{p+1} \cdots \varphi_{p+q}(\alpha) \hat{J}_{p+q}.$$

Corresponding to the set  $K_m$ , there is the connected component  $\text{int}(*\beta_m) / \bigcup_{v_j > v_m} \text{int}(*\beta_j)$  lying on the right of the  $y$ -axis; we will denote this set by  $L_m$ , for convenience. Equation (7.11) implies that  $K_m$  and  $L_m$  have equal areas. Furthermore, there is a 1-1 correspondence between the bounded components of  $(c_p, 0) \times \mathbb{R}$  and of  $(0, c_{p+1}) \times \mathbb{R}$ , namely that given by the vertical ordering of the lines. It follows that the area operator for a component of  $(c_p, 0) \times \mathbb{R}$  which contains the interval  $[w'_m, z'_m]$  on its boundary is equal to the area operator for the component of  $(0, c_{p+1}) \times \mathbb{R}$  which contains  $[w_m, z_m]$  on its boundary (see the discussion before and after Eq. (7.11) for this notation), and similarly for the other intervals  $[w'_m, w'_j]$ ,  $[z'_j, z'_m]$  and  $[z'_j, w'_i]$ .

For each value of  $m$ , we have already chosen  $d_m$  components of  $(c_p, 0) \times \mathbb{R}$ . Let us pick one of these components, and call its area operator  $E'_m$ . Then  $E'_m$  is also the area operator of the corresponding component of  $(0, c_{p+1}) \times \mathbb{R}$ , and by construction this component belongs to  $L_m$ . Since  $L_m$  is connected, we can again apply Corollary 4.8 and Theorem 6.8. We get a new Bralic map over  $(0, c_{p+1})$

$$\tilde{\tilde{\varphi}}_{p+1}(\alpha) = \exp \left[ -\alpha \sum_{m=1}^k |L_m| E'_m \right]$$

and we then have

$$\langle \tilde{W}(\alpha; \varepsilon) \rangle = \hat{J}_0 \hat{J}_1 \cdots \hat{J}_{p-1} \tilde{\varphi}_p(1 - \alpha) \tilde{\tilde{\varphi}}_{p+1}(\alpha) \hat{J}_{p+1} \cdots \hat{J}_{p+q-1} \hat{J}_{p+q}.$$

Finally we make use of the relation (7.11). In the proof of Theorem 6.8, we established the result by moving operators  $\exp[-|K|E]$  from one interval to another. The end result was that each connected component was represented by one single exponential operator  $\exp[-A_s E^{(s)}]$ , which contained the total area  $A_s$  of the component. However, we could now undo this process, and split up the operator  $\exp[-A_s E^{(s)}]$  among different intervals, in each of which it would appear as another exponential operator. But there is no reason why the coefficients of the

area operators in these exponentials should be the same as in the original expression. Indeed as long as their sum is exactly  $A_s$ , so that they can be recombined into the single area operator  $\exp[-A_s E^{(s)}]$ , these individual coefficients can be chosen freely.

Consider now the  $d_m$  components of  $(c_p, 0) \times \mathbb{R}$  which were chosen previously. Their corresponding components in  $(0, c_{p+1}) \times \mathbb{R}$  all belong to  $L_m$ , and their area operators are  $\{E_{m,i}\}$ . Therefore if we define

$$\tilde{\varphi}_{p+1}(\alpha) = \exp \left[ -\alpha \sum_{m=1}^k \sum_{i=1}^{d_m} |K_{m,i}| E_{m,i} \right],$$

it follows from (7.11) and the previous discussion that

$$\langle \tilde{W}(\alpha; \varepsilon) \rangle = \hat{J}_0 \cdots \hat{J}_{p-1} \tilde{\varphi}_p(1-\alpha) \tilde{\varphi}_{p+1}(\alpha) \hat{J}_{p+1} \cdots \hat{J}_{p+q}.$$

This expression is independent of  $\alpha$ , for any  $\varepsilon$  sufficiently small. Therefore  $\langle \tilde{W}(\alpha) \rangle$  is also independent of  $\alpha$ .

## APPENDIX

We state below three lemmas which imply the results used in Definition 7.14, Theorem 7.26 and Lemma 7.29. In each case,  $W$  is an admissible collection of closed curves, and  $p, p'$  are points in the image  $W(I)$ . We will assume that  $p, p'$  are not intersection points of  $W$ , and that the tangent vectors are defined at these points.

**LEMMA A1.** *Let  $T$  be the tangent vector at  $p$ , and let  $v$  be any other vector satisfying  $\text{sgn}(T, v) \neq 0$ . Then there is a  $C^1$  curve  $\gamma$  satisfying  $\gamma(0) = p$ ,  $\gamma'(0) = v$  and  $\gamma(t) \in W(I)^c$  for all  $0 < t \leq 1$ .*

*Remark.* The proof of this lemma is illustrated in Fig. A1. Since all curves in  $W$  are continuous, and since the tangent vector is defined at  $p$ , there is a

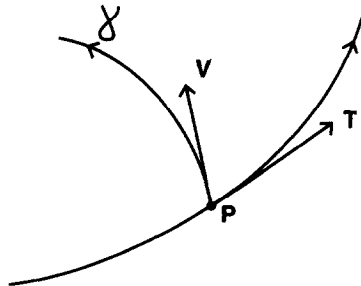


FIGURE A1

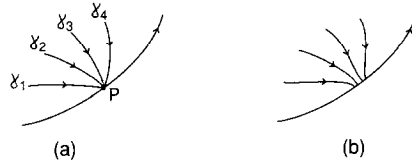


FIGURE A2

neighbourhood of  $p$  in which the curve looks like the picture in Fig. A1. Lemma A1 implies the existence of the curves  $\{\Gamma_m\}$  in Definition 7.14.

**LEMMA A2.** *Let  $\{\gamma_1, \dots, \gamma_m\}$  be  $C^1$  curves satisfying  $\gamma_i(1) = p$ ,  $\text{sgn}(\gamma'_i(1), T) > 0$  for all  $1 \leq i \leq m$ , where  $T$  is the tangent vector at  $p$ . We assume furthermore that the curves are disjoint, and do not intersect  $W(I)$ , except at the point  $p$ . Then the curves can be deformed in a neighborhood of  $p$  so that they become completely disjoint, with distinct endpoints on the curve near  $p$ .*

*Remark.* The lemma is illustrated in Fig. A2, which also serves as the proof. Note that the condition  $\text{sgn}(\gamma'_i(1), T) > 0$ ,  $1 \leq i \leq m$ , can be replaced by the condition  $\text{sgn}(\gamma'_i(1), T) < 0$ , for  $1 \leq i \leq m$ .

**LEMMA A3.** *Let  $\Gamma$  be a  $C^1$  curve with  $\Gamma(0) = p$  and  $\Gamma(1) = p'$ , whose image is otherwise disjoint from  $W(I)$ . Let  $N_\varepsilon(\Gamma)$  be the  $\varepsilon$ -neighborhood of the set  $\Gamma(I)$ . Let  $\{q_1, \dots, q_m\}$  be distinct points lying on the same curve to which  $p$  belongs, and within a distance  $\varepsilon$  from  $p$ . Then there exist  $C^1$  curves  $\{\gamma_1, \dots, \gamma_m\}$  satisfying  $\gamma_i(0) = q_i$ ,  $\gamma_i(1) = p'$  for  $1 \leq i \leq m$ . Furthermore, for  $\varepsilon$  sufficiently small, these curves may be chosen so that  $\gamma_i(t)$  belongs to  $N_\varepsilon(\Gamma) \cap W(I)^c$  for  $0 < t < 1$ , each  $1 \leq i \leq m$ , and so that the curves are disjoint except at  $p'$ .*

*Remark.* Once again the proof of this lemma is illustrated in Fig. A3.

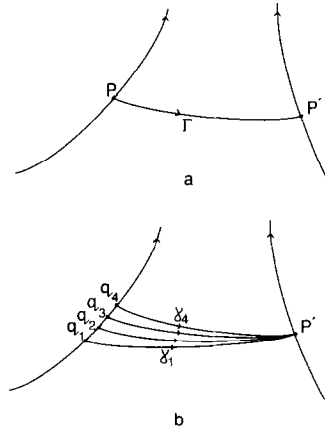


FIGURE A3

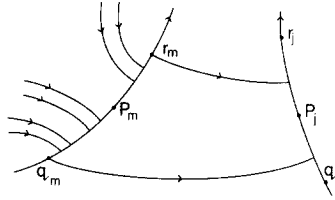


FIGURE A4

We can now address the proof of Lemma 7.29. We have a tree  $\Gamma$  on the set  $W(I)$ , connecting the gates by  $C^1$  curves. At each gate  $p_m$ , there are a number of incoming curves and one outgoing curve. By Lemma A2, the incoming curves can be deformed so that their endpoints are disjoint. Since they originate from different gates, these deformed curves are now entirely disjoint. Let  $\Gamma_i^*$  be one such curve, so that  $\Gamma_i^*(0) = p_i$ . The points  $q_i, r_i$  lie on the same curve as  $p_i$ , within a distance  $\varepsilon$ . Therefore by Lemmas A3 and A2, we can find two disjoint curves within the  $\varepsilon$ -neighborhood of  $\Gamma_i^*(I)$  which connect  $q_i$  and  $r_i$  to points on the curve near  $p_m$ . This procedure is followed for every incoming curve at  $p_m$ , in such a way that all the resulting curves are disjoint, and so that their endpoints all lie between  $q_m$  and  $r_m$ . The resulting situation at  $p_m$  is illustrated in Figure A4. It is clear that the ordering of the endpoints between  $q_m$  and  $r_m$  is determined by the coefficients  $\{\delta(j, m)\}$ . Furthermore, suppose that  $\bar{m} = j$ , so that the two outgoing curves shown in Fig. A4, whose initial points are  $q_m$  and  $r_m$ , go through the region  $R_j$  and terminate on the curve near  $p_j$ , between  $q_j$  and  $r_j$ . Then by Lemmas A3 and A2 all the incoming curves near  $p_m$  can be extended through the region  $R_j$  so that their endpoints lie on the curve near  $p_j$ , and so that they remain disjoint. The result is illustrated in Fig. A5. By iterating this construction at every gate in the tree  $\Gamma$ , we generate  $2k$  disjoint curves which connect the points  $\{q_m, r_m\}$  to distinct points on the  $y$ -axis near  $p_0$ . These are the curves  $\{\rho_m, \tau_m\}$ .

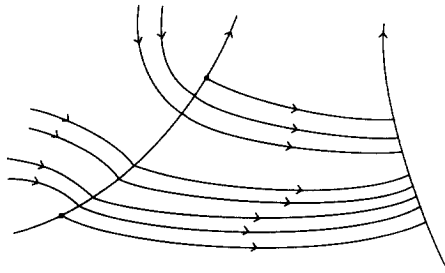


FIGURE A5

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